Introduction to Laser Trimming Technology

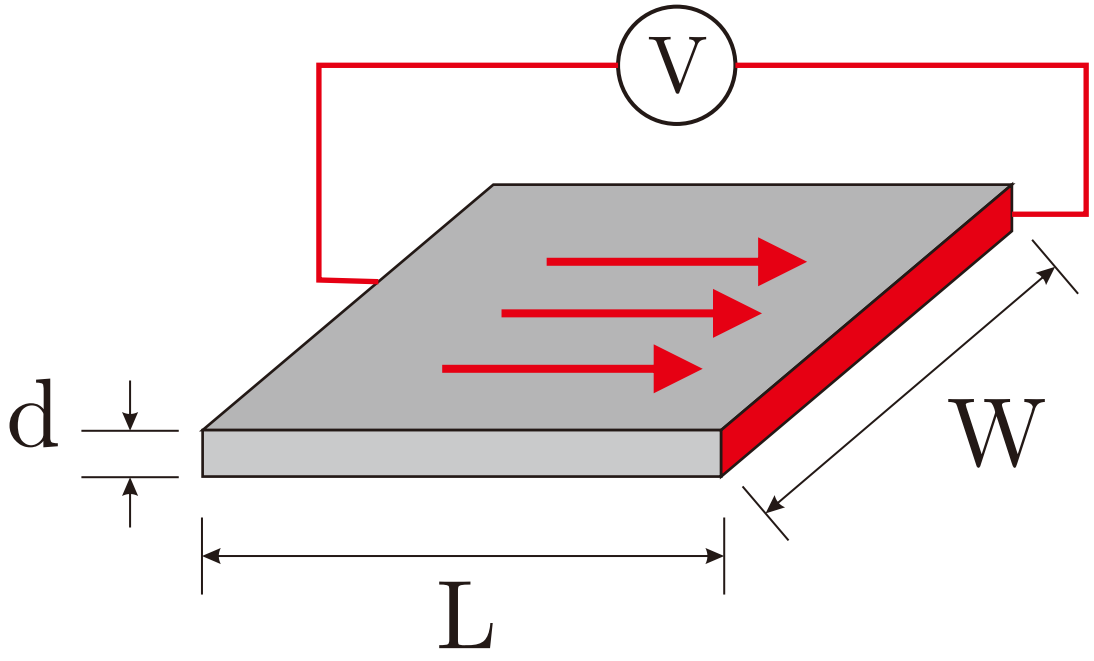
雷射修阻技術簡介

Chih-Han Lin 林致翰

HyPro 均鈺科技股份有限公司

clin@hypro.com.tw TEL:(03)3756109

Sheet resistance & Ohm's Law



cross sectional area:

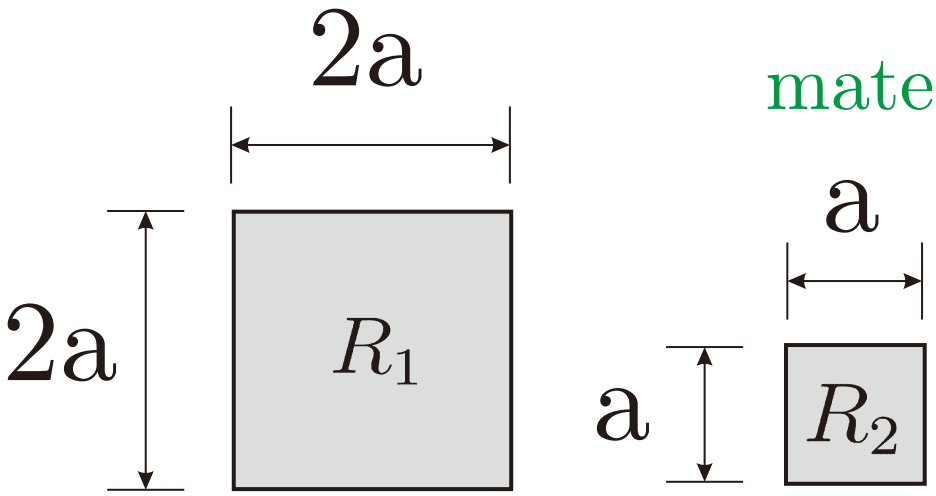
$A = d \times W$

resistance:

$$R = \rho\left(\frac{L}{A}\right) = \frac{\rho}{d}\left(\frac{L}{W}\right)$$

sheet resistance, $\Omega/\square$ — material, thickness

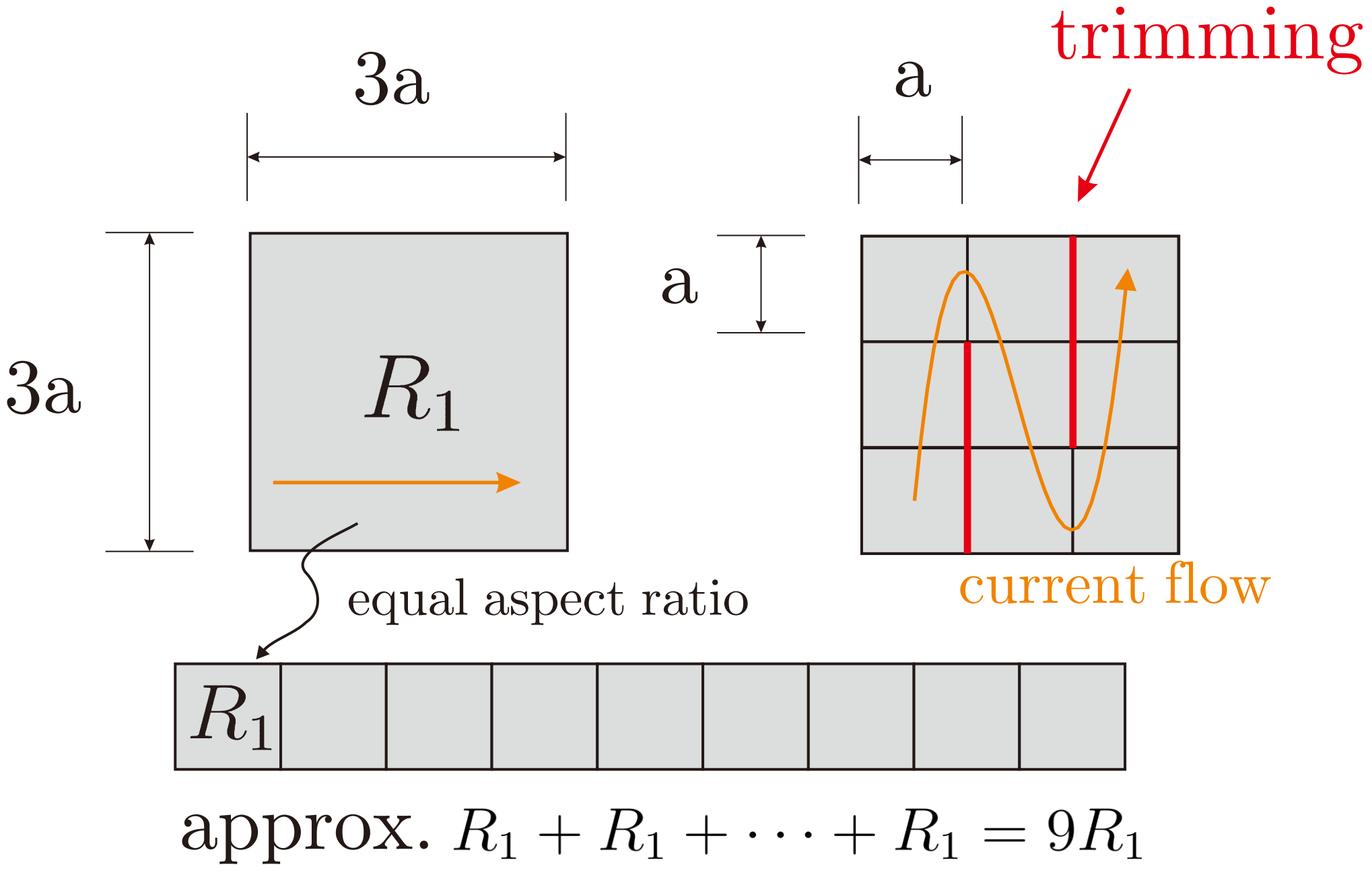
aspect ratio — geometric



from top view: $R_1 = R_2$

(with equal thickness)

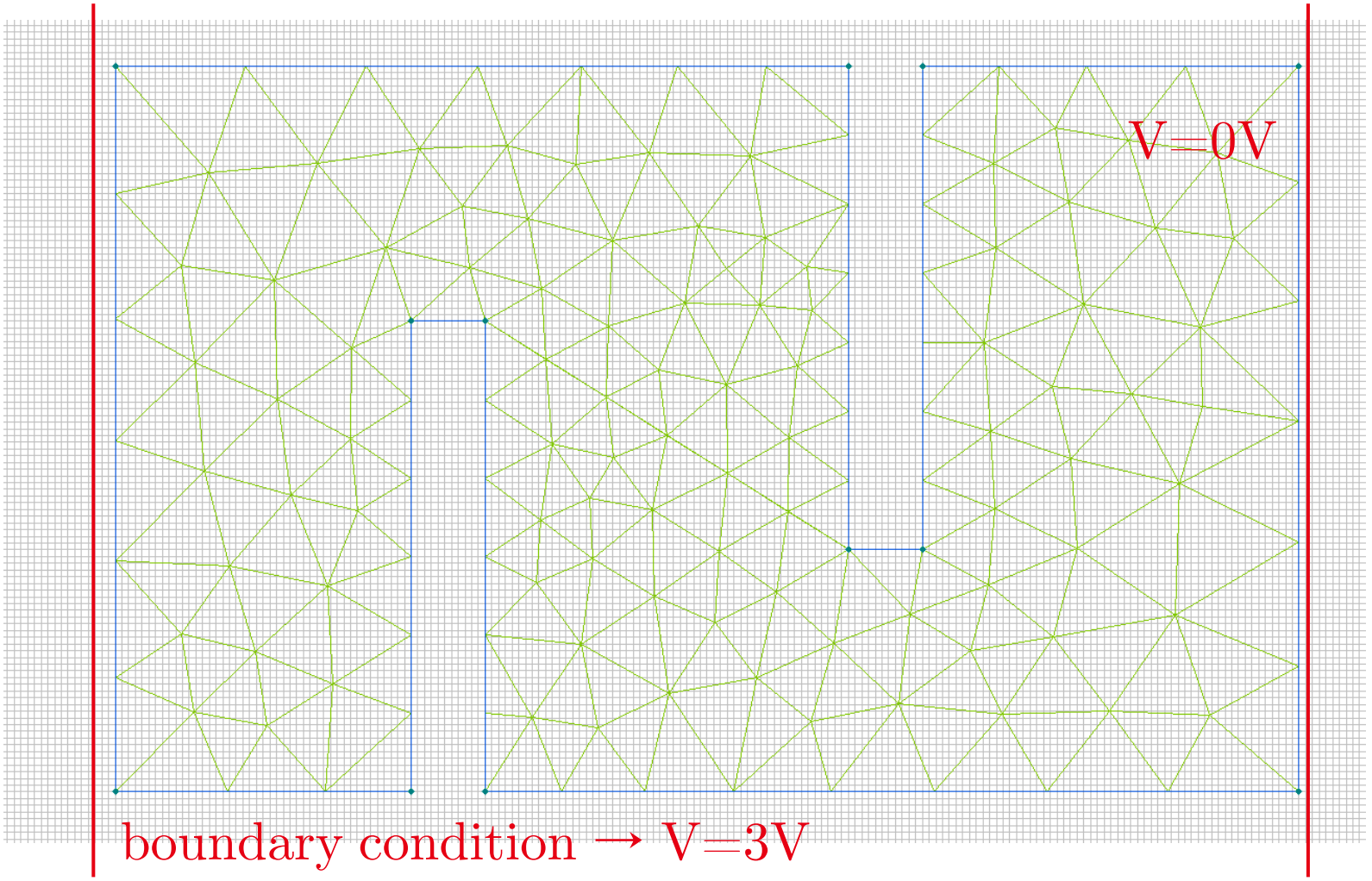
Concept of Resistance-Trimming



approx. $R_1 + R_1 + \cdots + R_1 = 9R_1$

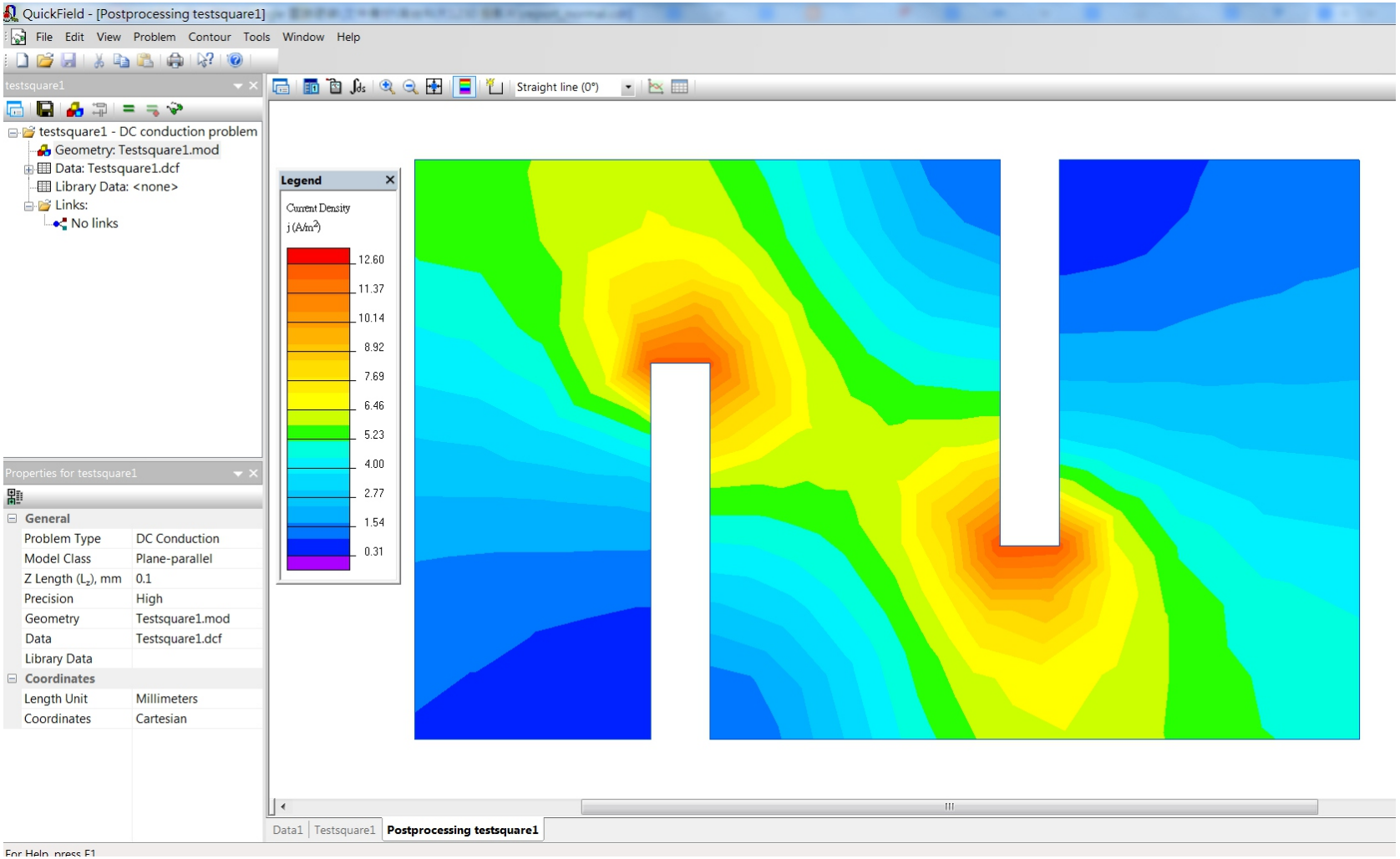
Electromagnetic Analysis of Resistance-Trimming

Finite element analysis (QuickField student version)



software download: http://www.quickfield.com/

Electromagnetic Analysis of Resistance-Trimming



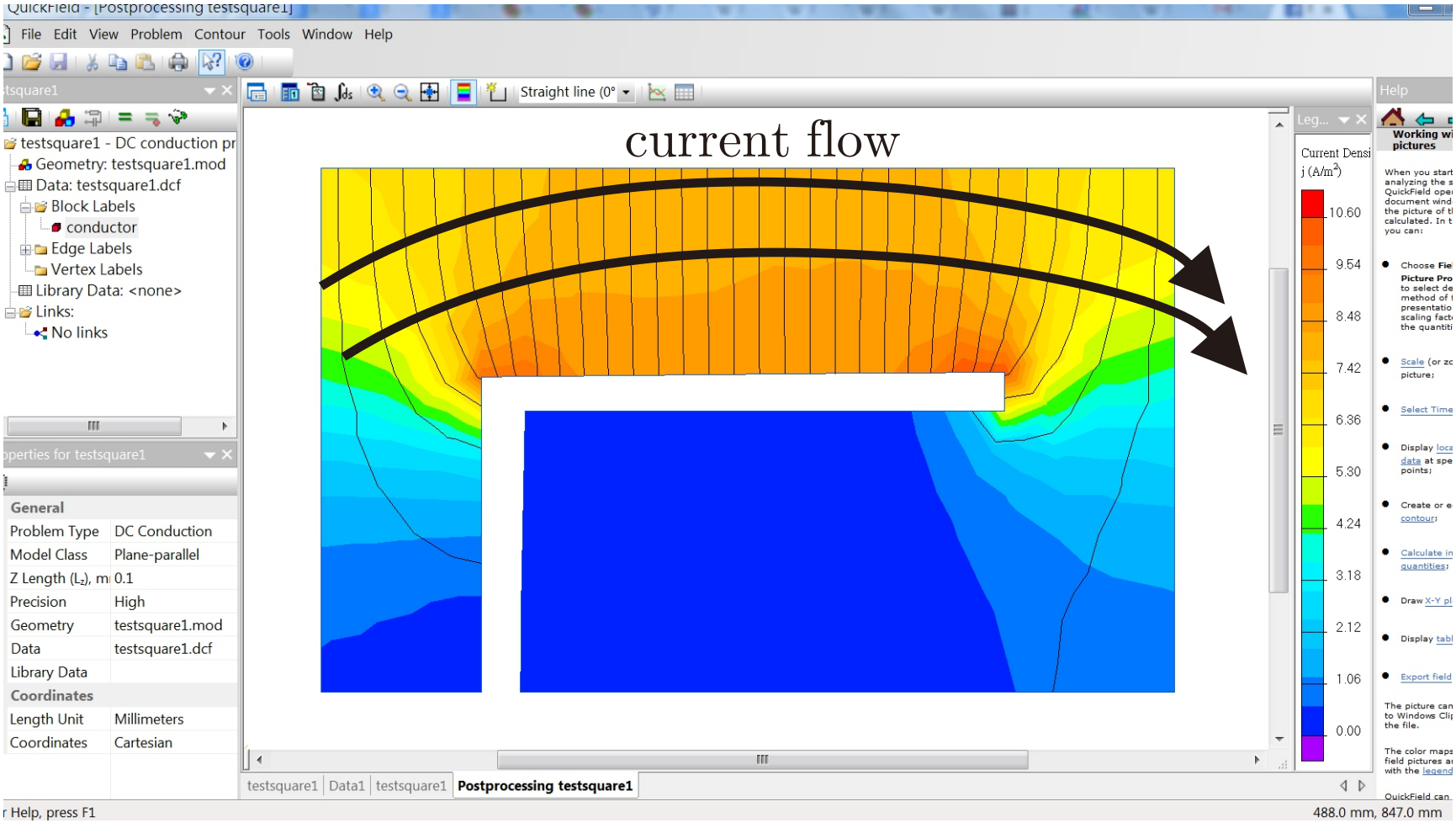
Simulation result of surface current density

Electromagnetic Analysis of Resistance-Trimming



Simulation result of surface potential distribution

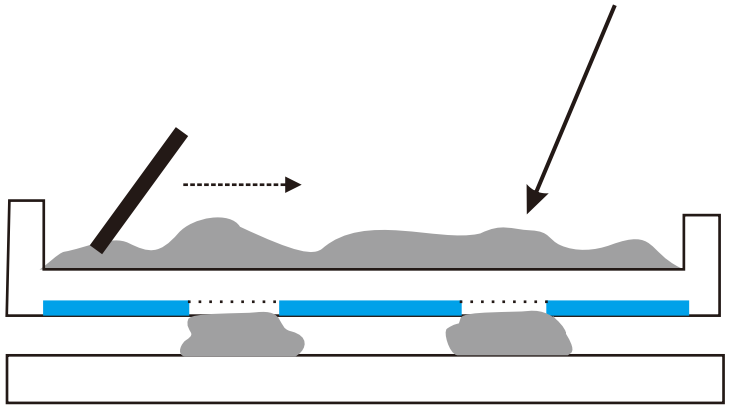
Electromagnetic Analysis of Resistance-Trimming



another simulation of typical L-shape cut

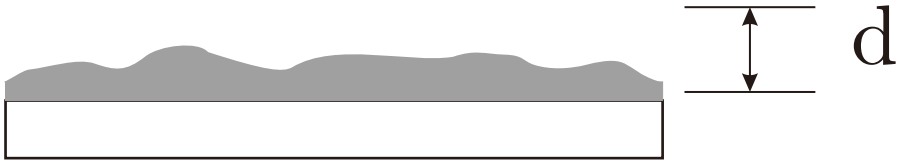
Why do we need trimming?

resistor paste based on oxides of ruthenium, iridium, zirconium and rhenium



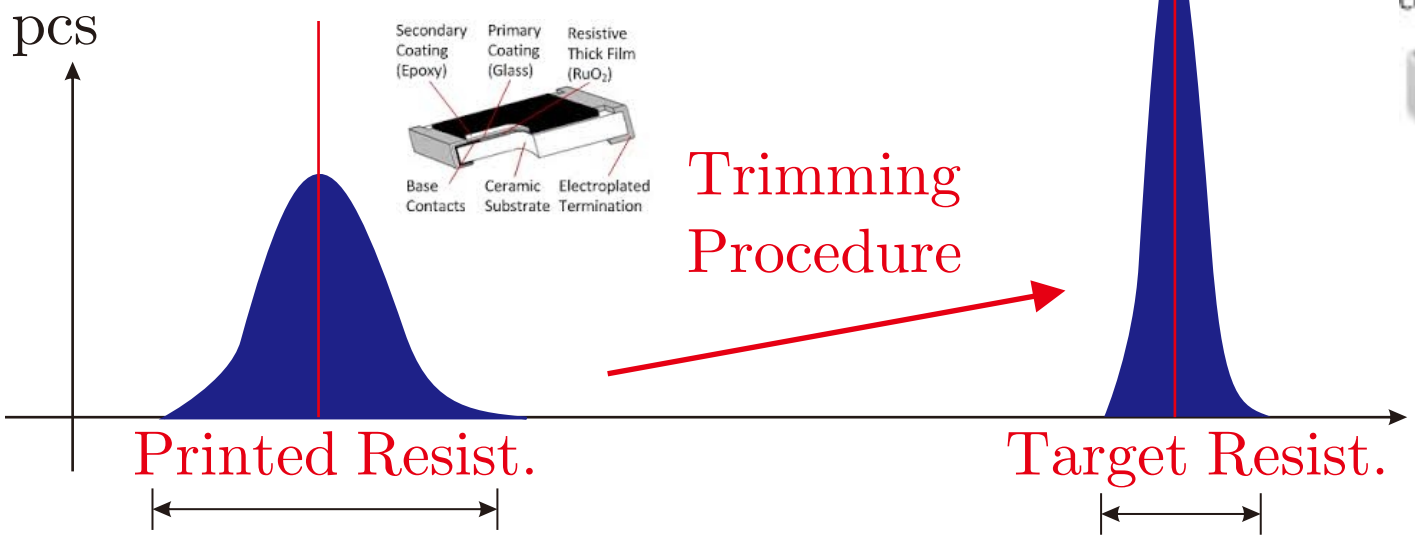
silk screen printing

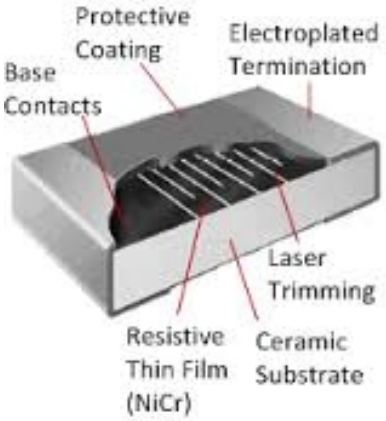
IrO2, ZrO2 and RuO2 etc.



ceramic baseboard

baking



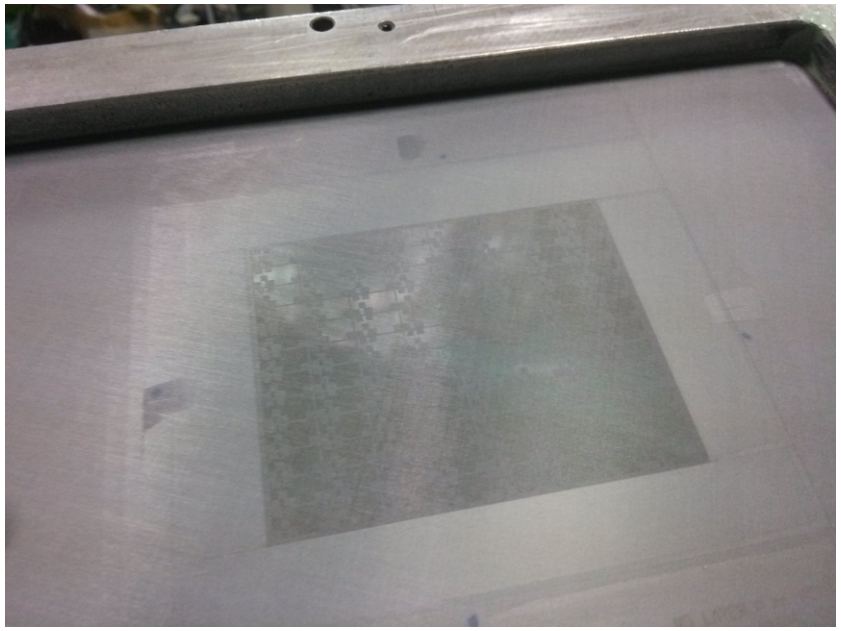


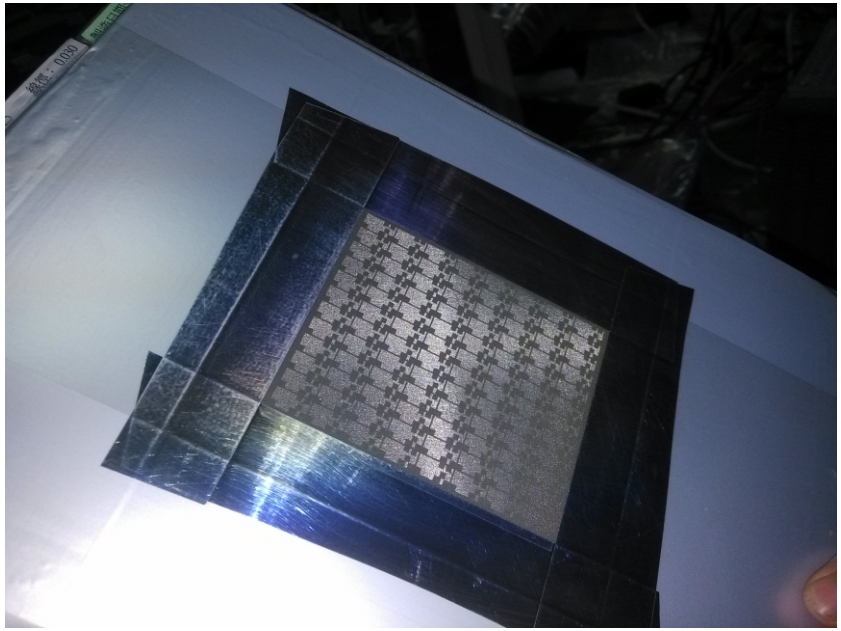
±10% error

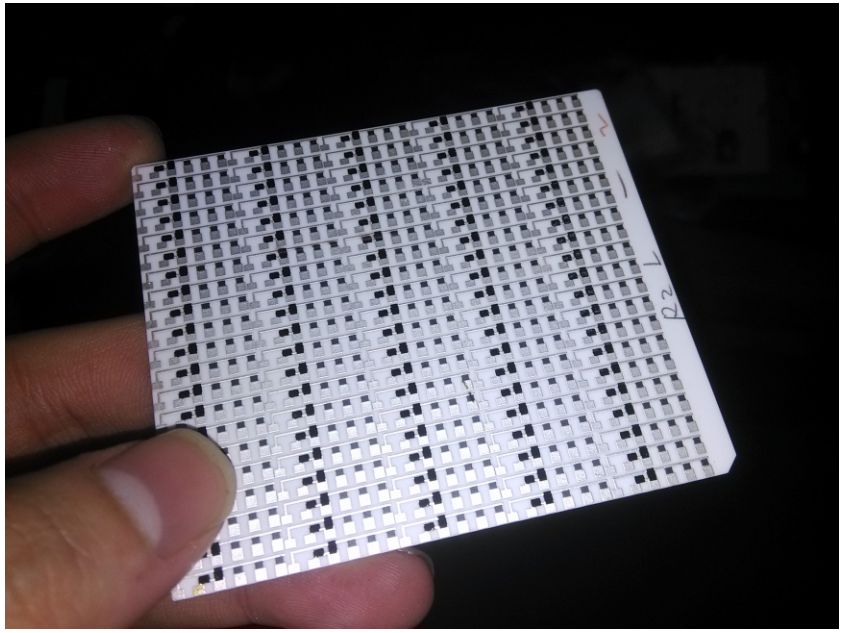
±1% to ±0.1% error

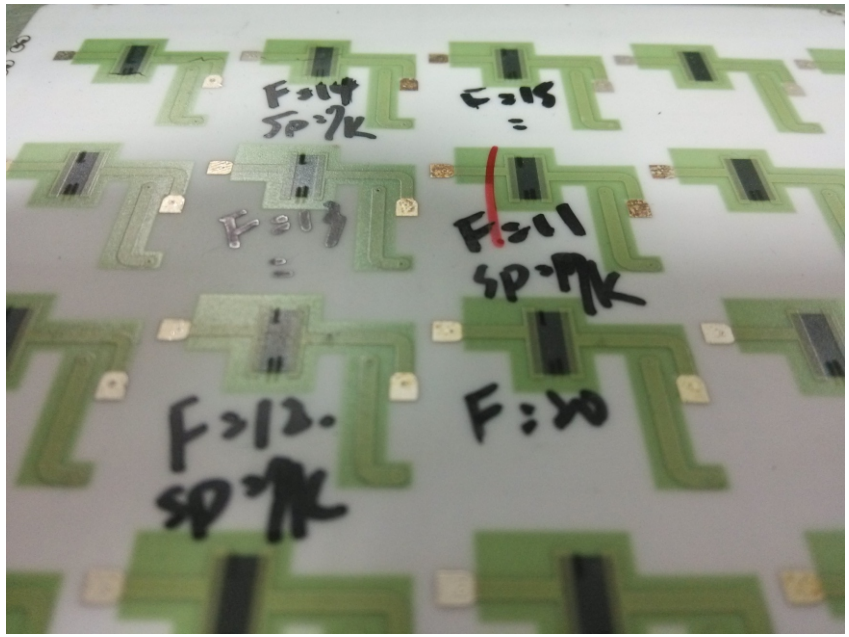
pic. URL: www.resistorguide.com

Thick film printed ceramic substrate





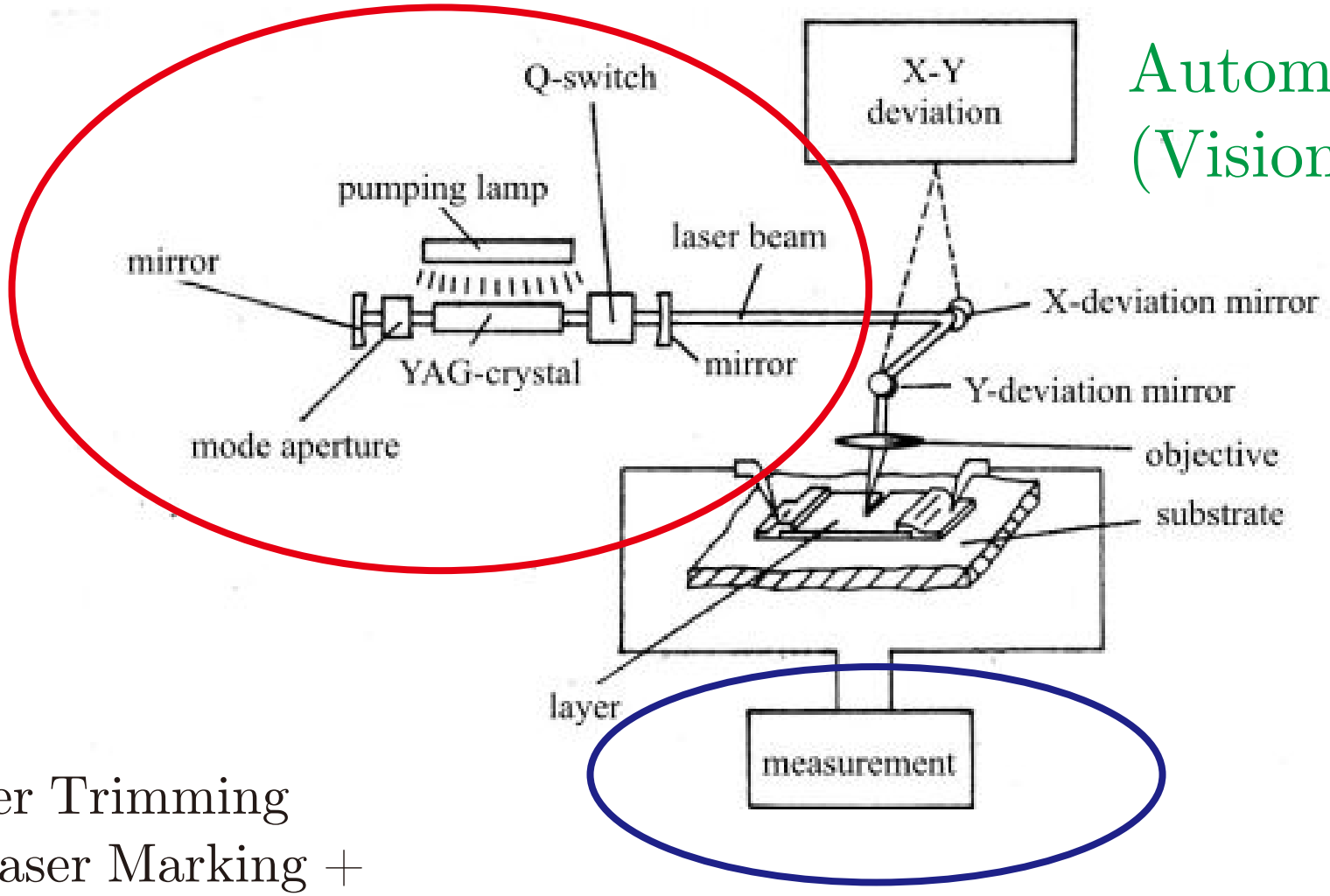




clin@hypro.com.tw TEL:(03)3756109 Chih-Han Lin 林致翰 HyPro 均鈺科技股份有限公司

Schematic structure of a laser trimming system

Laser machining beam optics



Automatic Control
(Vision & Motion)

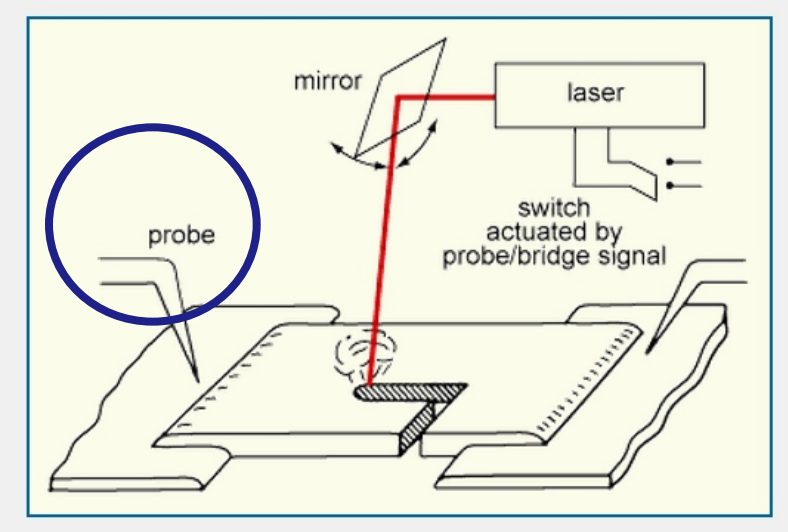
Laser Trimming
= Laser Marking +
measurements

Dynamic measurements

Schematic structure of a laser trimming system

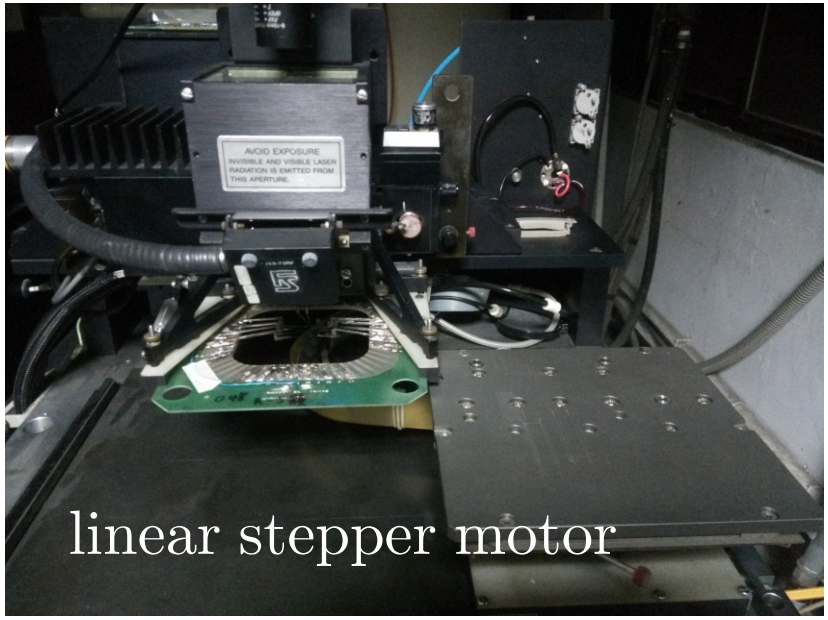


Schematic structure of a laser trimming system



http://www.ami.ac.uk/courses/topics/0255_tft/







Laser source & Beam characteristics

Q-switched Nd:YAG Laser (up to 10W)



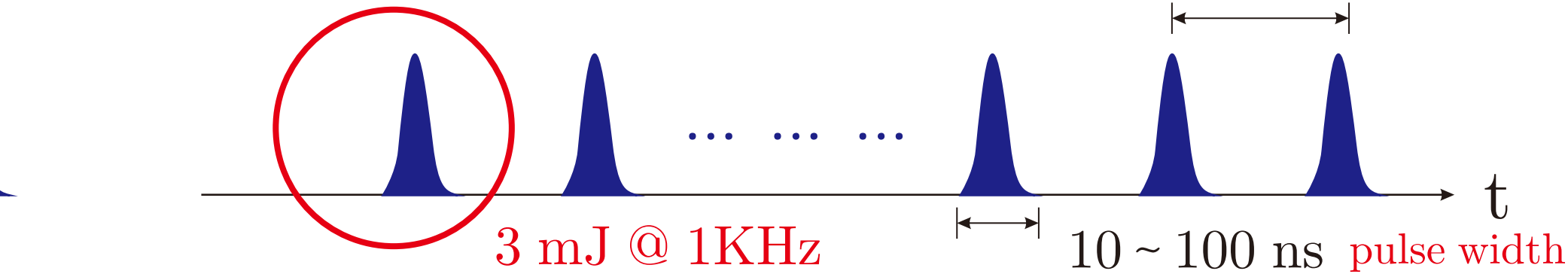
http://www.ipgphotonics.com/

Pulsed Q-switched fiber Laser (up to 40W, longer pulse width)

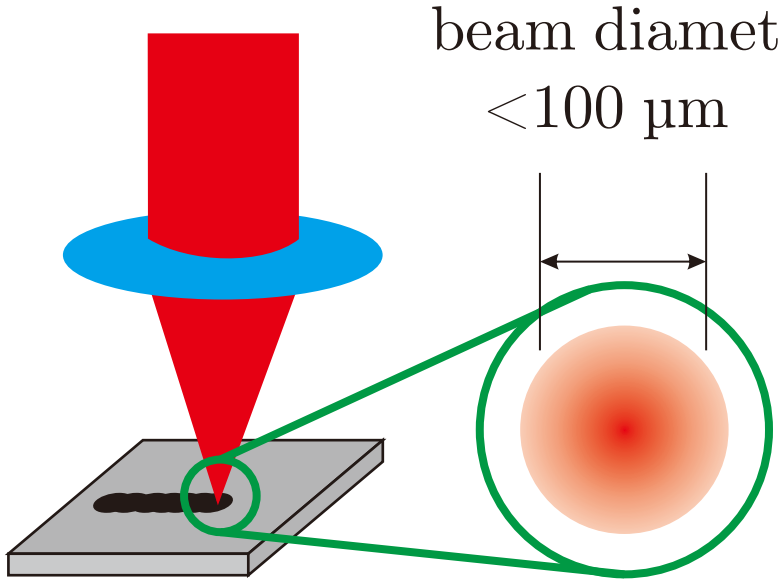


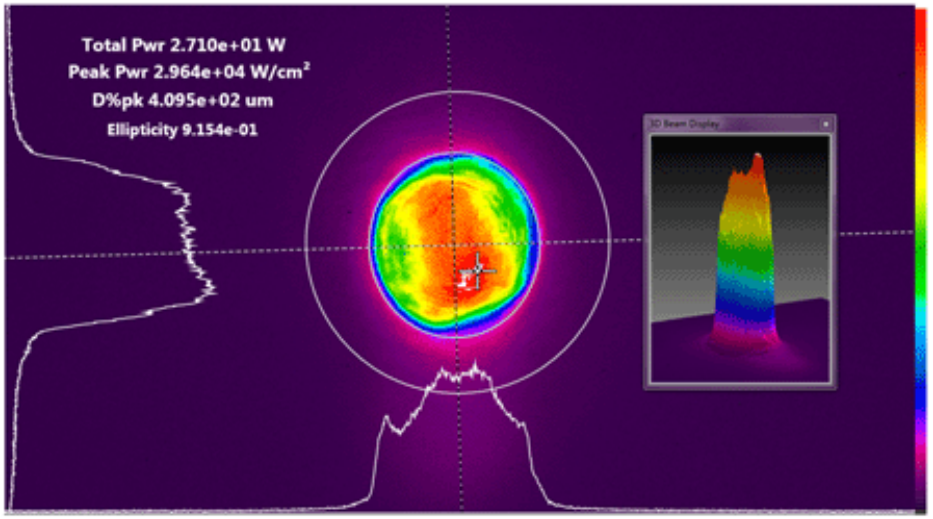
http://www.coherent.com/

$20 \sim 1000\ \mu s \rightarrow 1K \sim 50\ KHz$ repetition rate



Laser source & Beam characteristics





practical laser beam profile

www.ophiropt.com

3 mJ @ 1KHz, 50 ns (3W average power)

Peak Power:

$$\frac{3\ \text{mJ}}{50\ \text{ns}} = 60\ \text{kW}$$

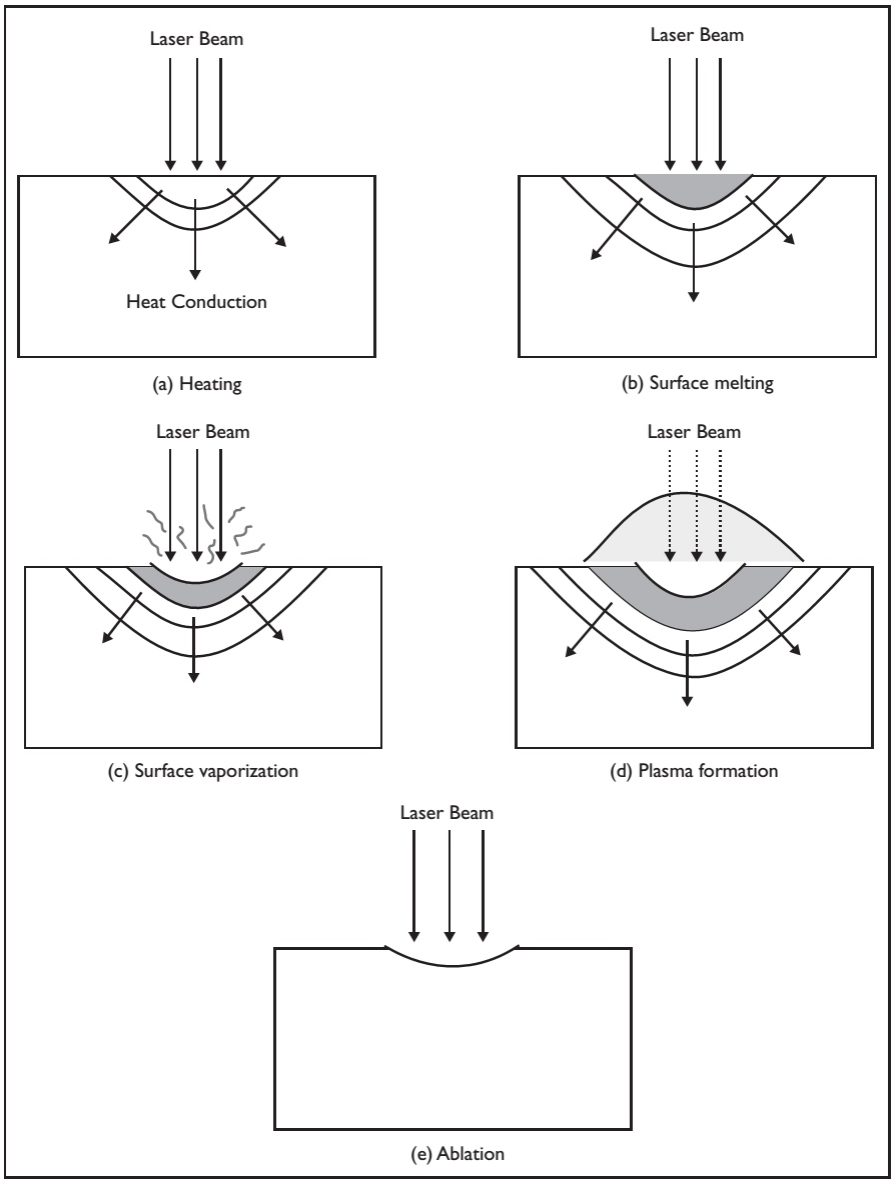
Peak Irradiance:

$$\frac{60\ \text{kW}}{(1/2)\pi(50\ \mu\text{m})^2} = 1.53\times10^9\ \frac{\text{W}}{\text{cm}^2}$$

Irradiance of Sun (focused)

$$\leftrightarrow\quad 10^4\frac{\text{W}}{\text{cm}^2}$$

Interaction between Light and Matter



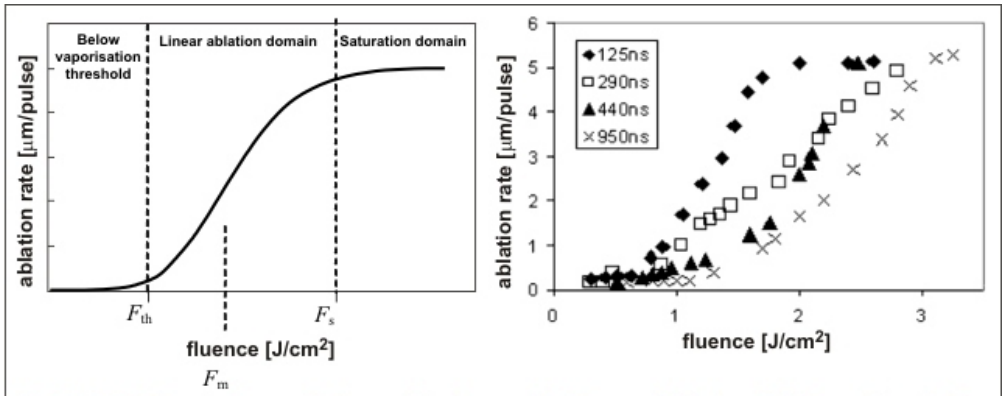


Fig. 2.1.9: Ablation rate for vaporisation-mediated removal. Left) general behaviour. Right) experimental data achieved for laser ablation of black crust standards using different pulse duration between 125-950 ns (SFR Nd:YAG laser).

© Siano, in: "Handbook on the Use of Lasers in Conservation and Conservation Science", 2008.

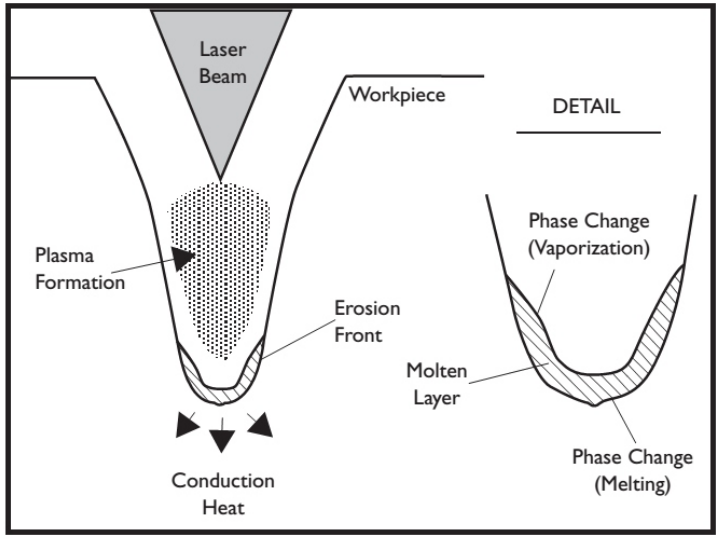
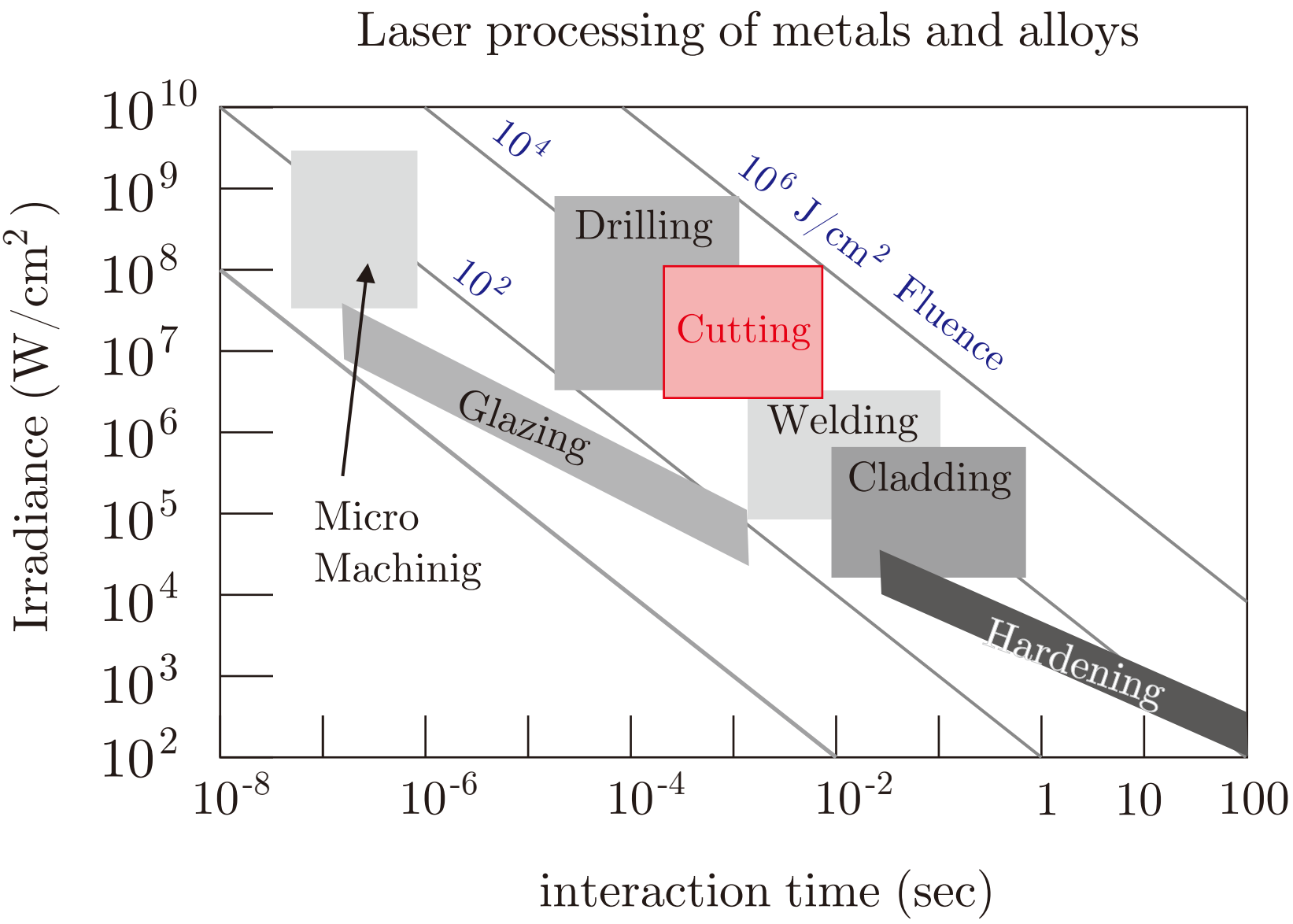


Figure 4.2 Schematic of laser drilling process (Reprinted from Chryssolouris, 1991 with kind permission of Springer Science + Business Media).

Laser Machining of Advanced Materials by Narendra B Dahotre, Anoop Samant, CRC Press , 2011

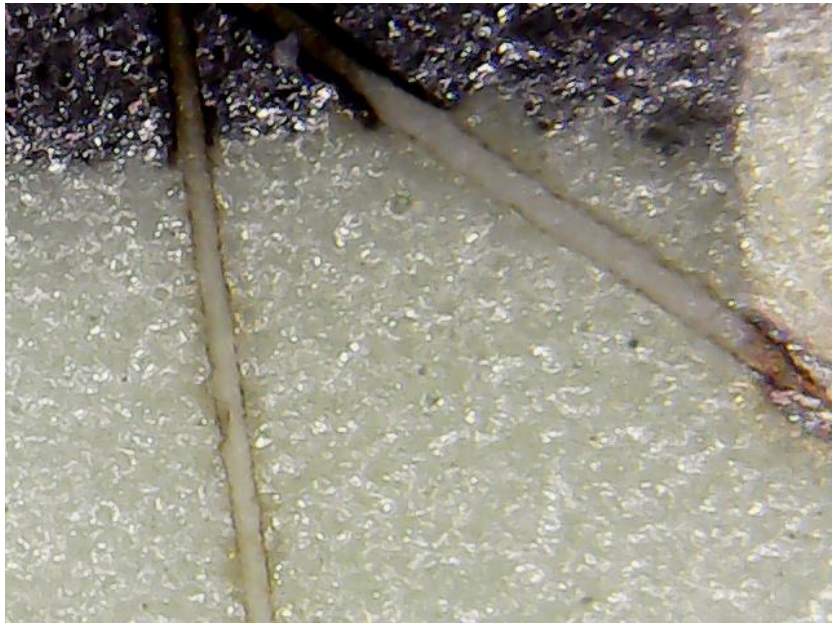
Interaction between Light and Matter



Interaction between Light and Matter









Find the best laser machining parameter

scanning speed (mm/sec)

Repetition rate (Hz)

focal spot size

Average Power (Watts)

Pulse width (ns)

Thermal Effect

Irradiance W/cm^2

Linear Energy Density $W/cm^2/mm$

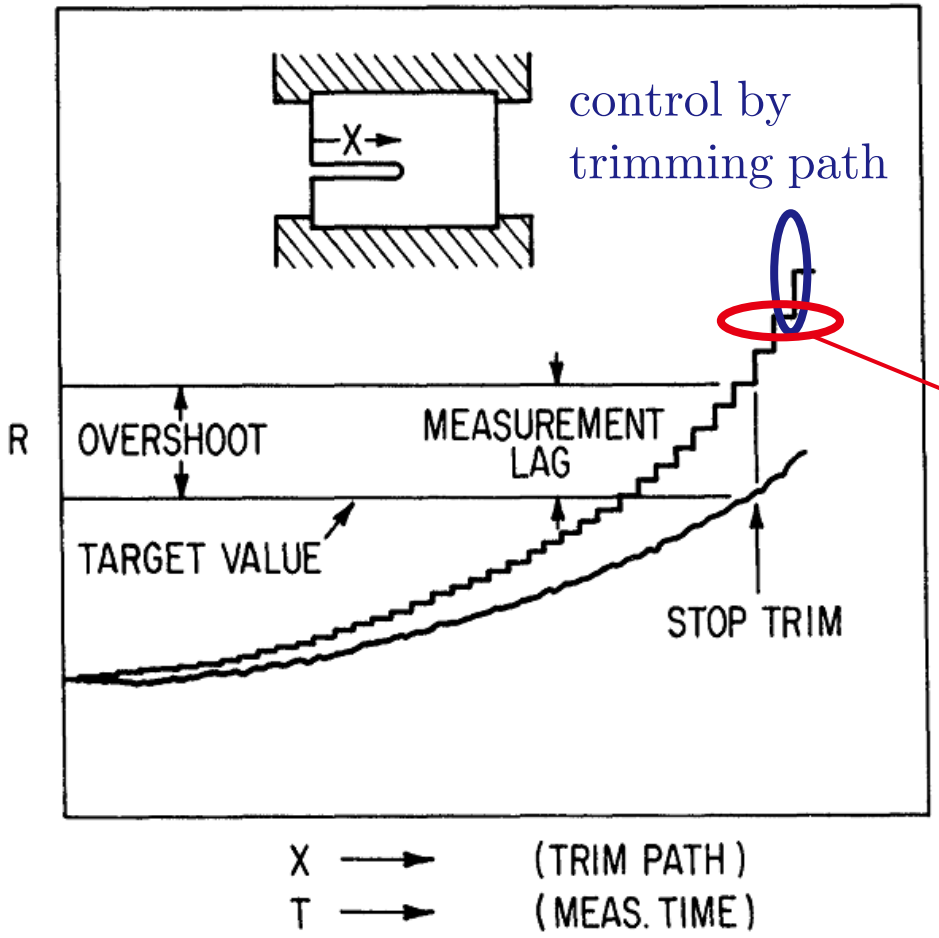
Fluence J/cm^2

Ablation Effect

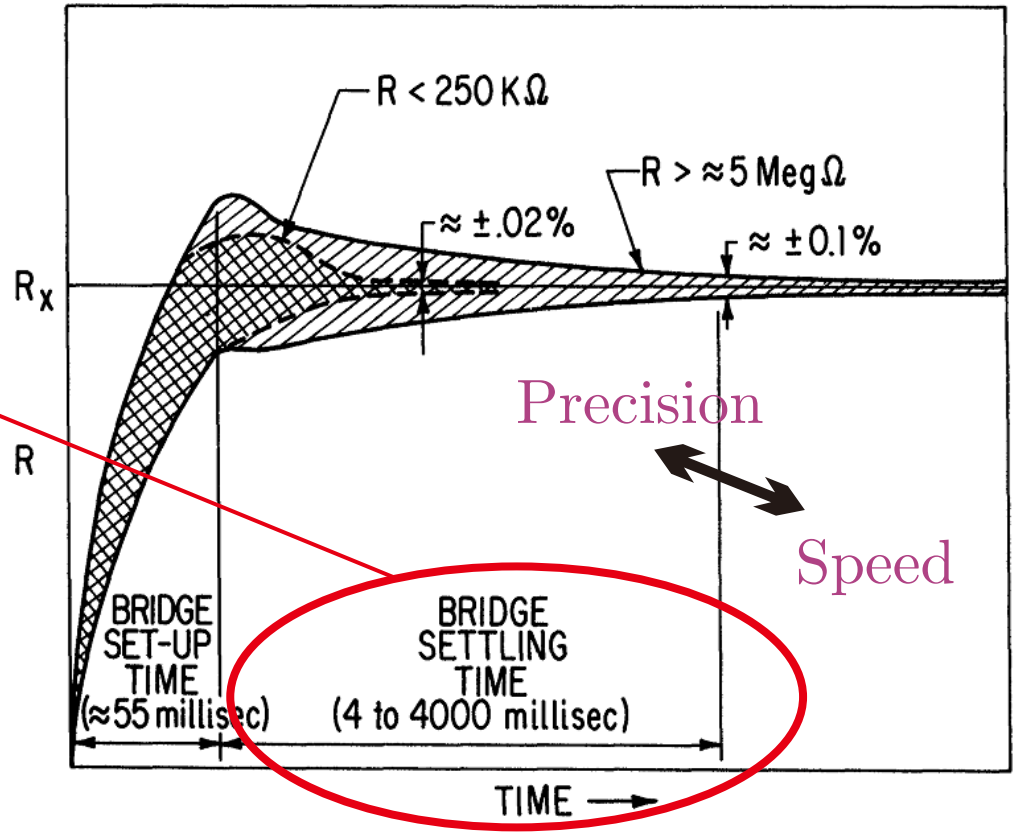
In general, we operate YAG laser at about 1 to 1.5 W, 3 KHz and 10 to 20 mm/sec (scanning speed) to achieve 1% tolerance resistors.

Fundamental of laser-trimmed film resistor

settling time of measurements



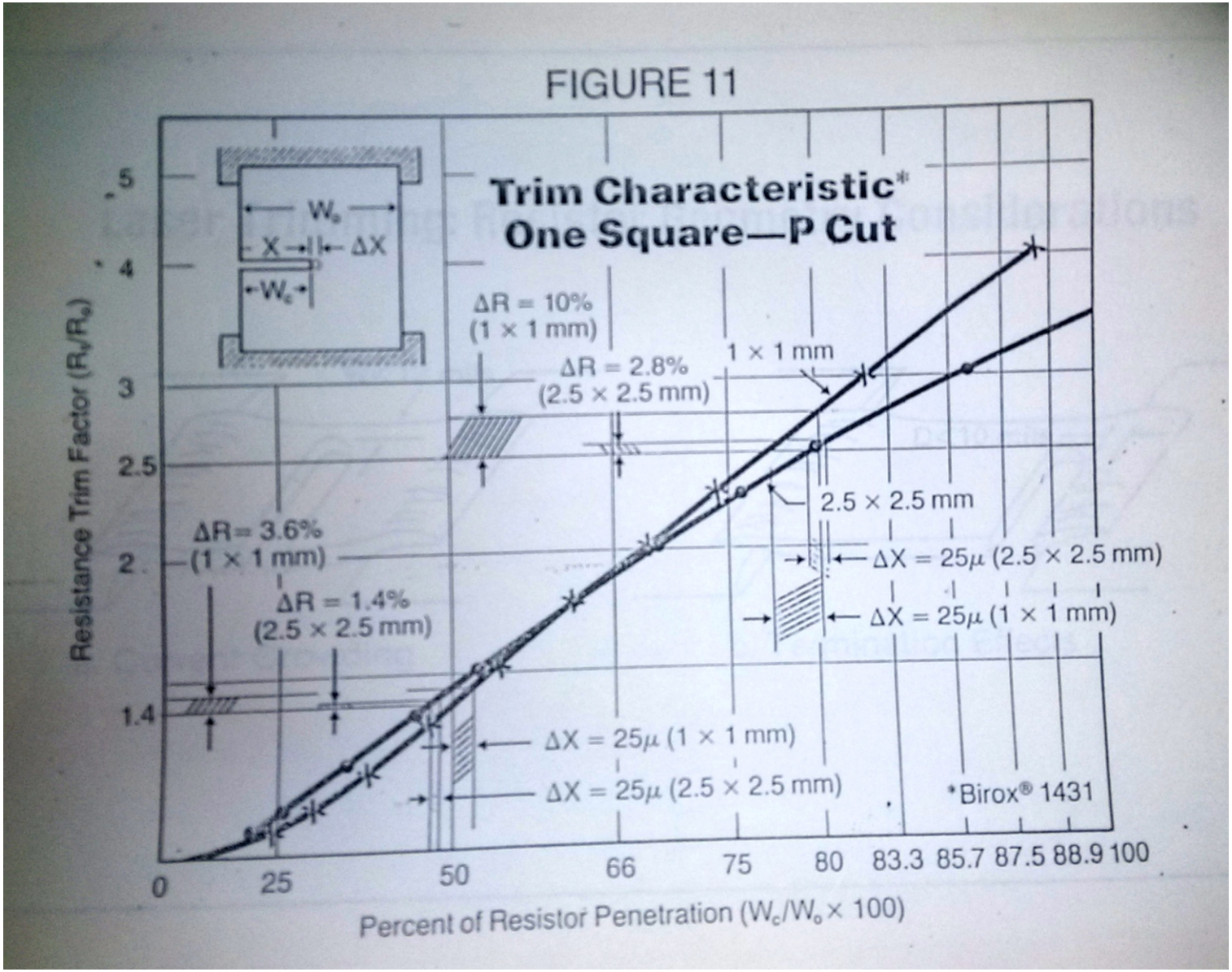
settling time of measurements



*ESI Mod. 25 YAG System. Auto Zero Mode.

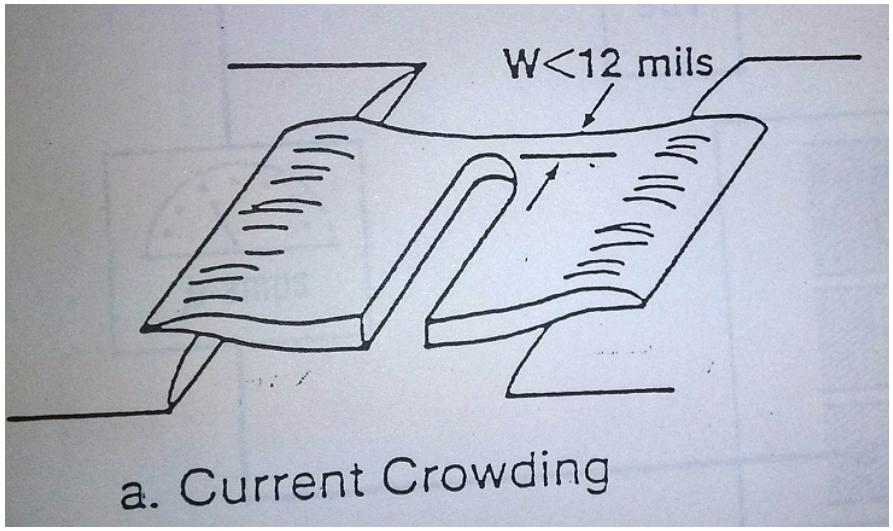
Factor affecting laser-trim stability of thick film resistors, R.E Cote & R. C. Headley, 1977

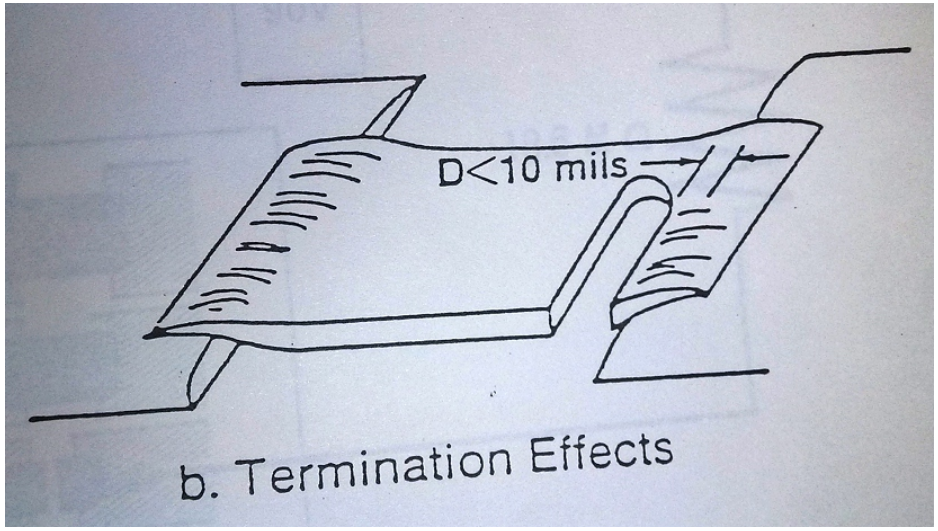
Fundamental of laser-trimmed film resistor



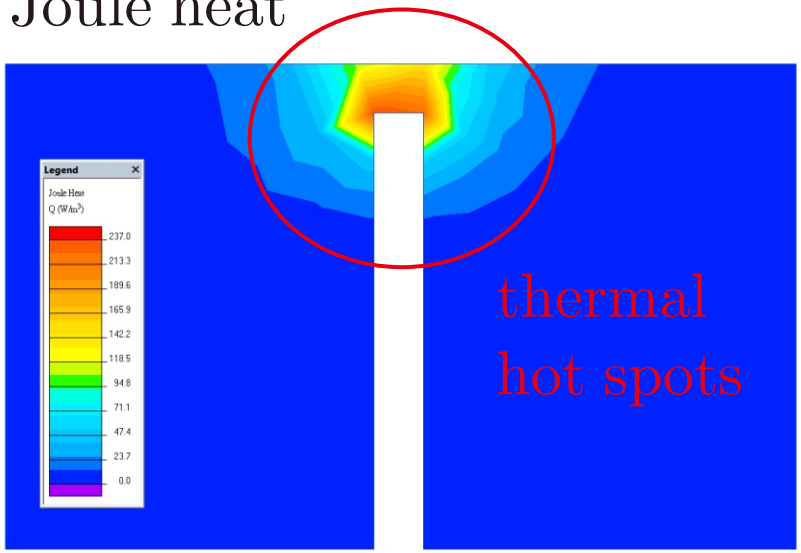
The effect of high speed laser trimming on accuracy and stability of thick film resistors, R.E Cote, 1981

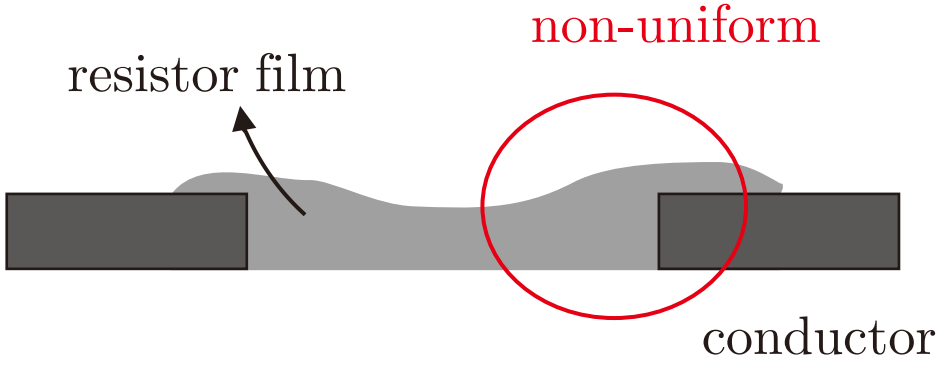
Geometry consideration of trimming path



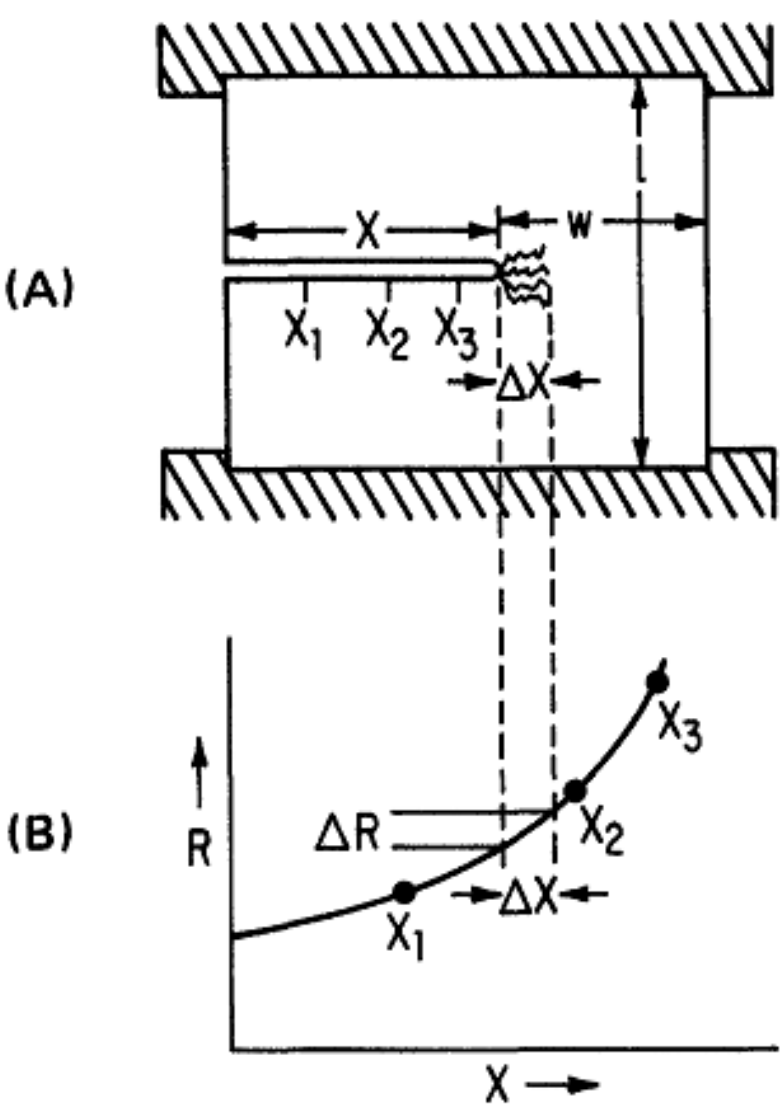


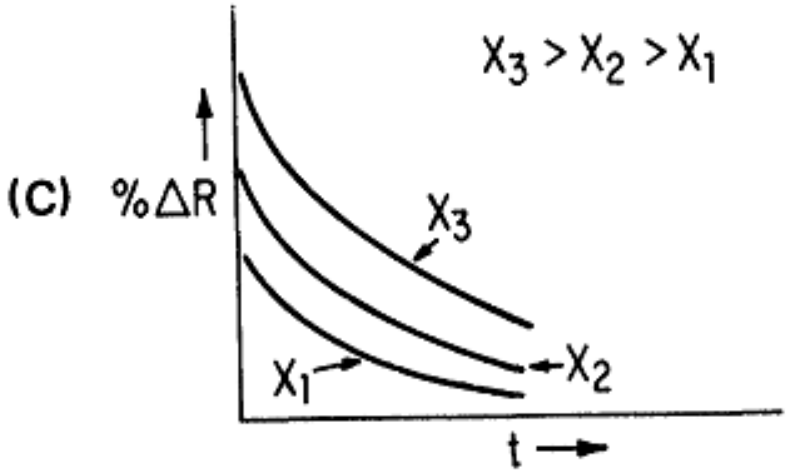
Joule heat

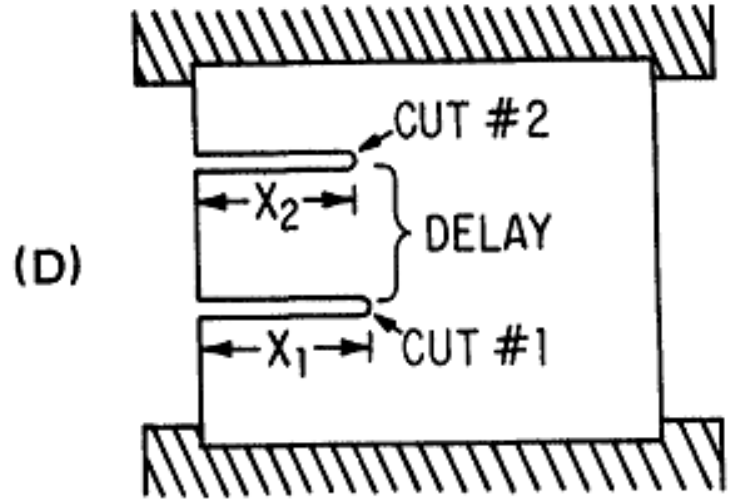




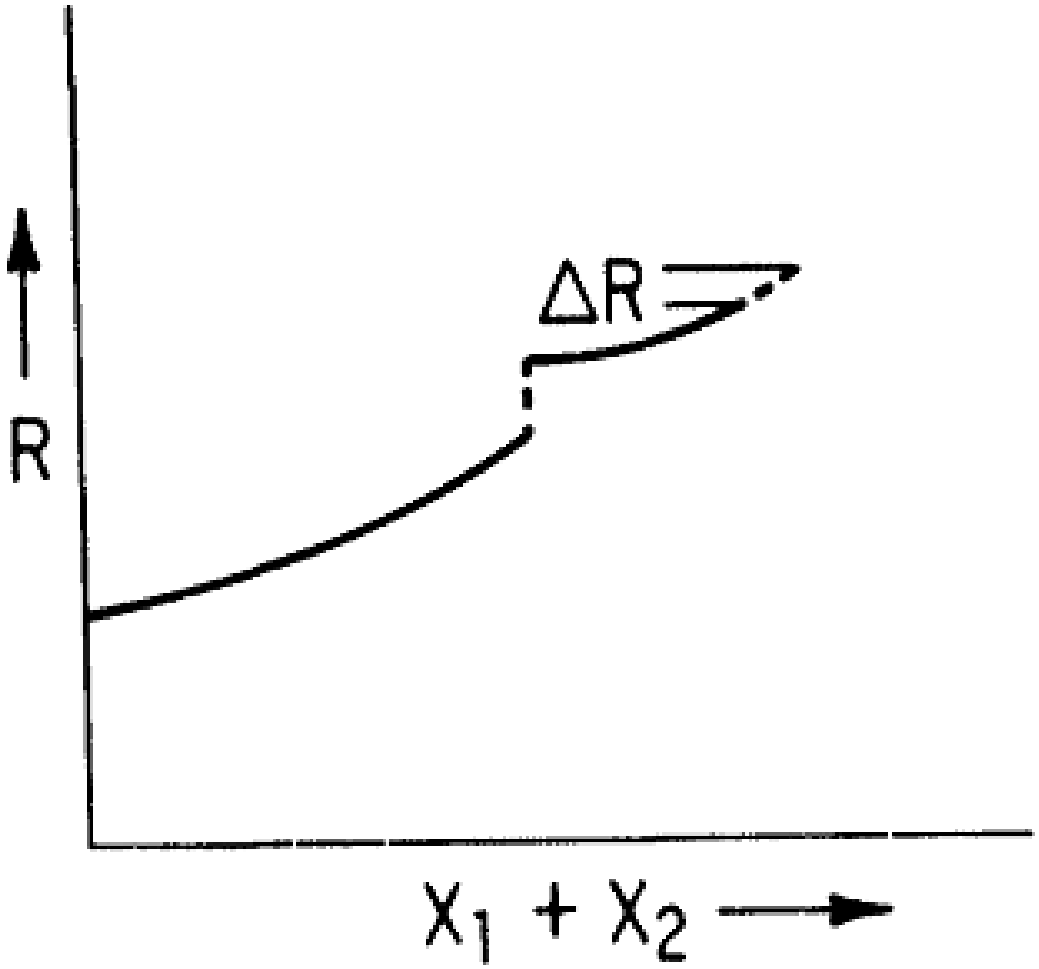
Fundamental of laser-trimmed film resistor





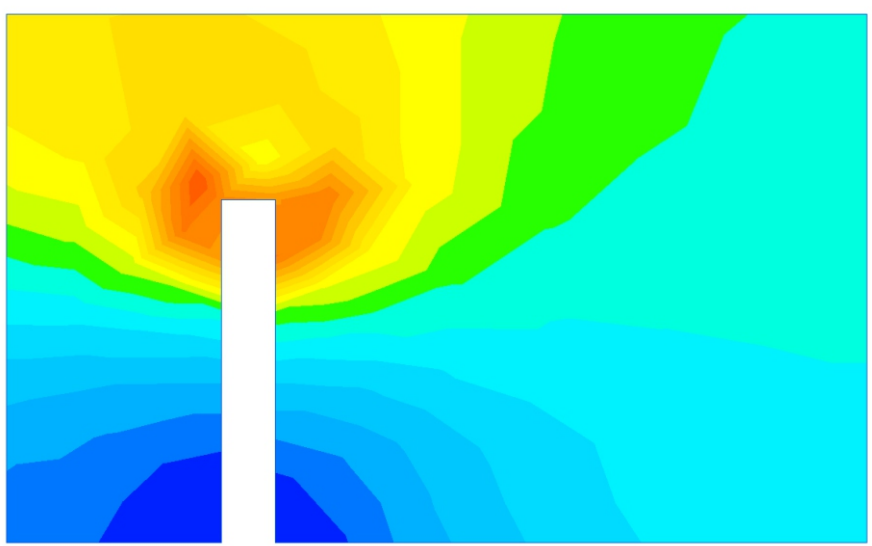


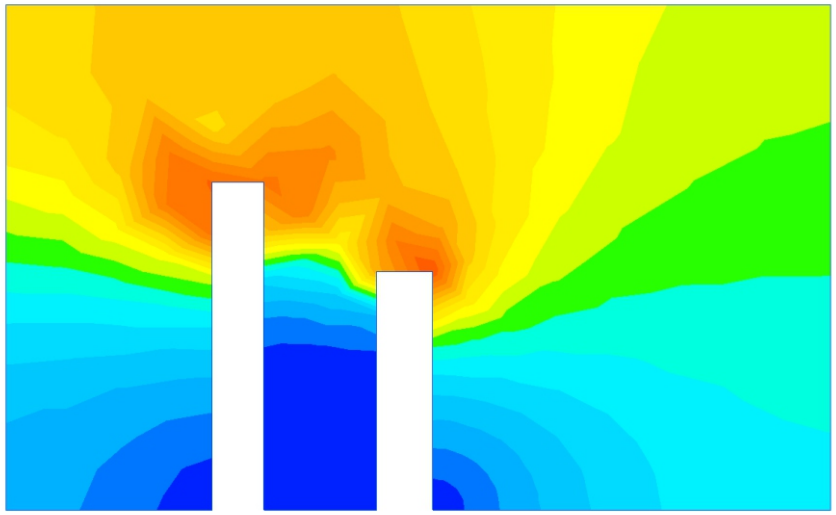
Plunge cut → Double cut

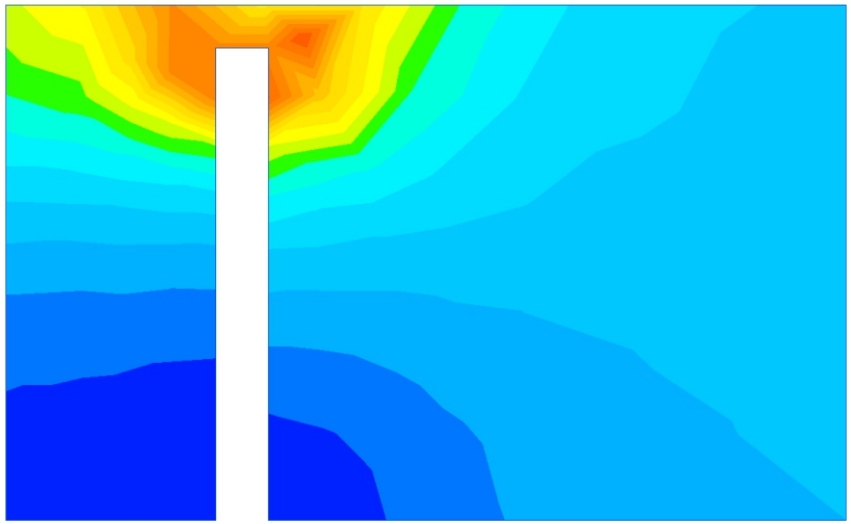


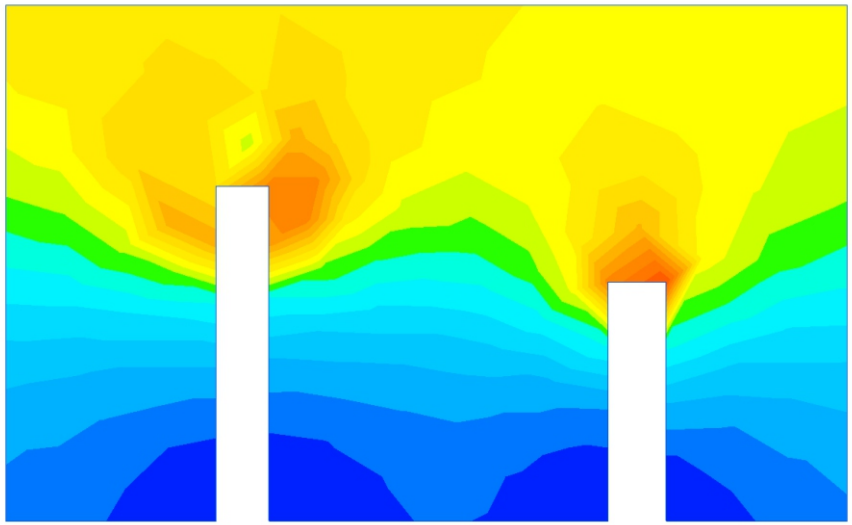
Finite Element Analysis of Double cut

Simulation result of surface current density



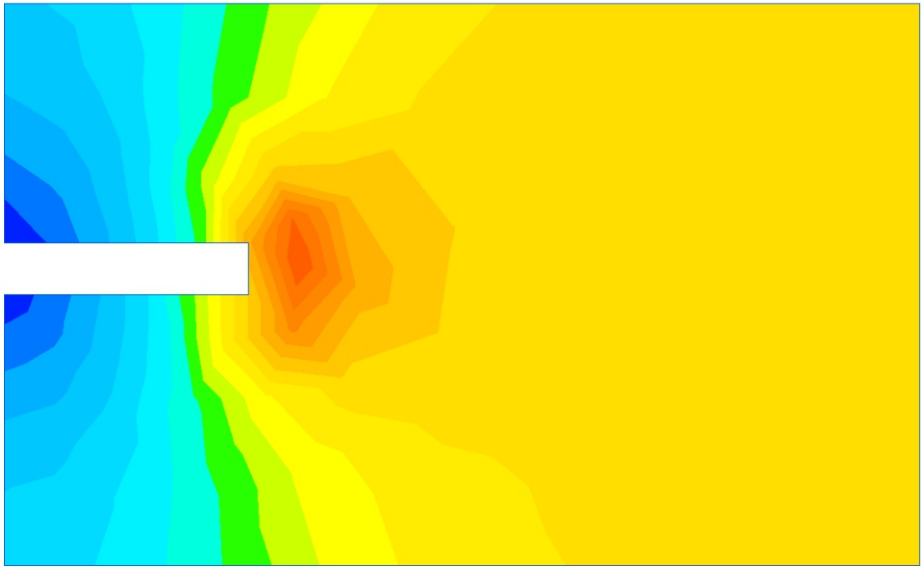


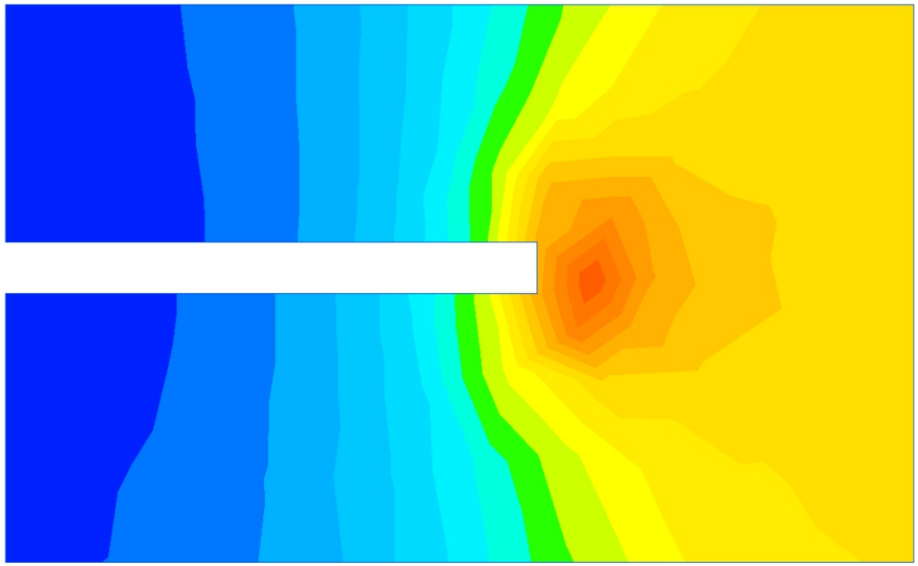


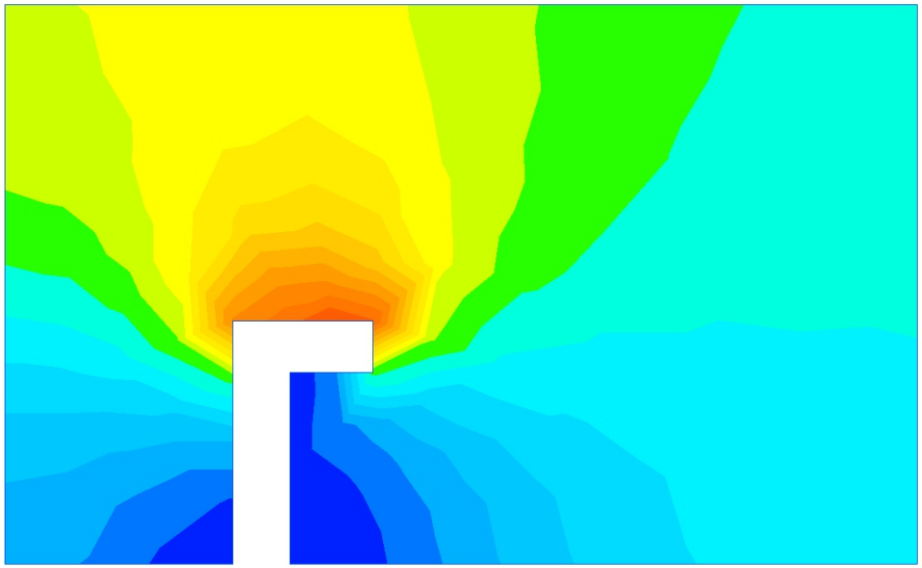


Finite Element Analysis of L-shape cut

Simulation result of surface current density









Finite Element Analysis of L-shape cut

Simulation result of surface potential distribution

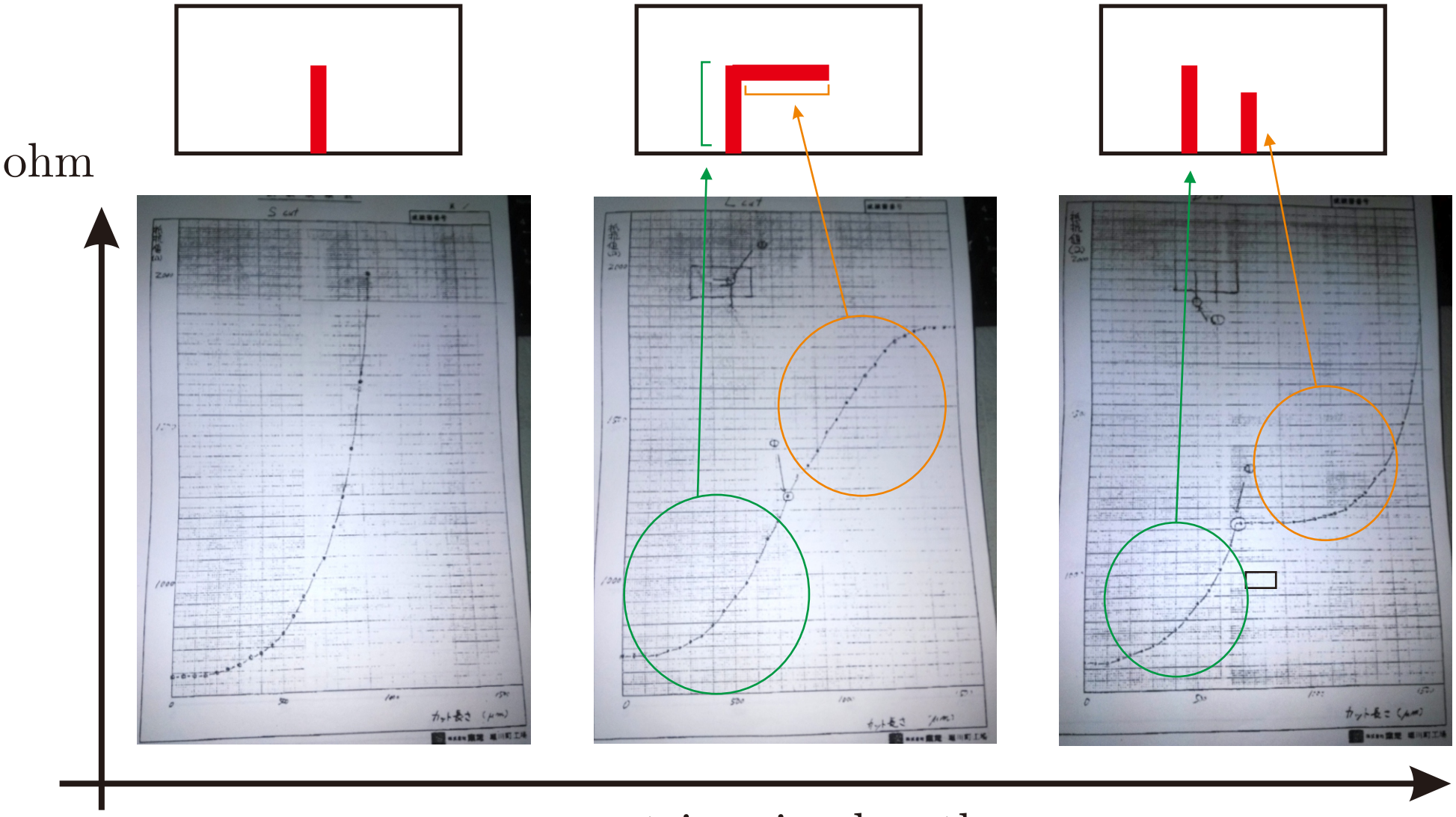


Trimming characteristic of different tragedy

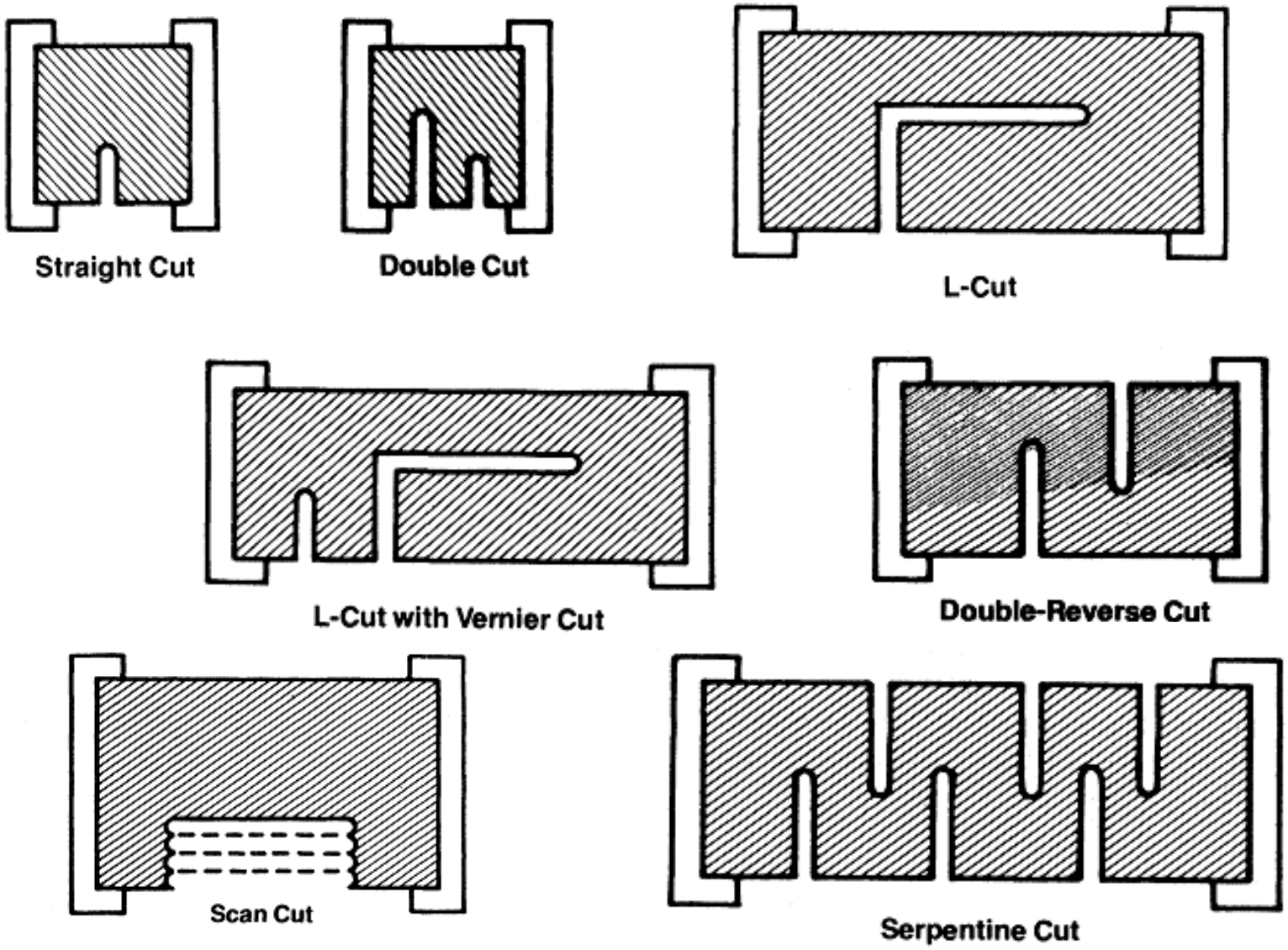
Plunge Cut

L-shape Cut

Double Cut



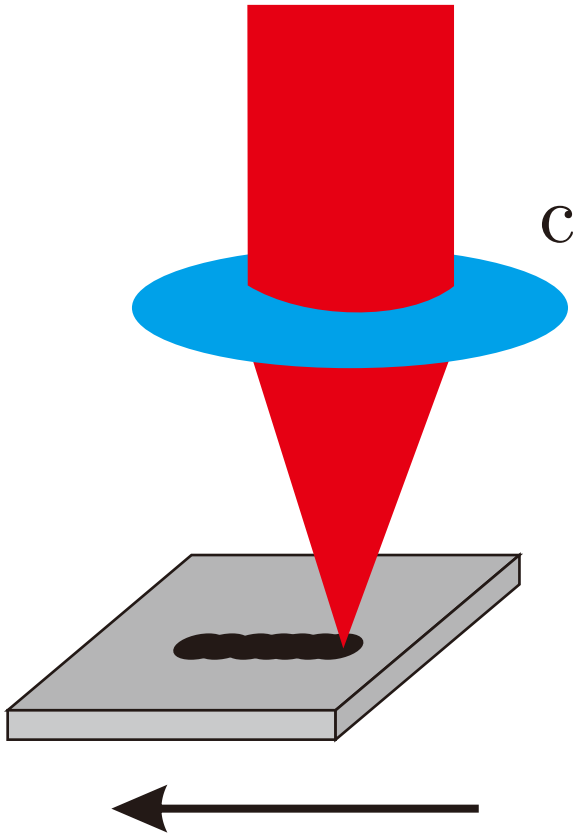
Trimming characteristic of different tragedy



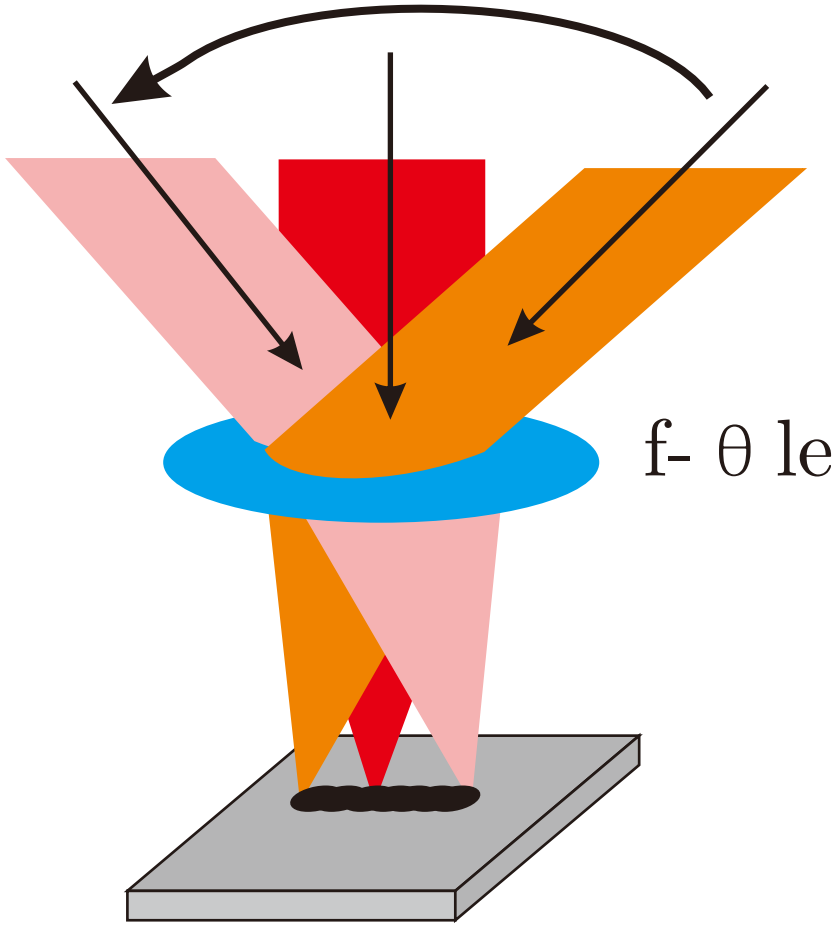
Thick-film resostpr trimming principle, techniques and recommendations, R.E Cote, 1981

Machining beam delivery & focal spot position

machining beam



conventional lens

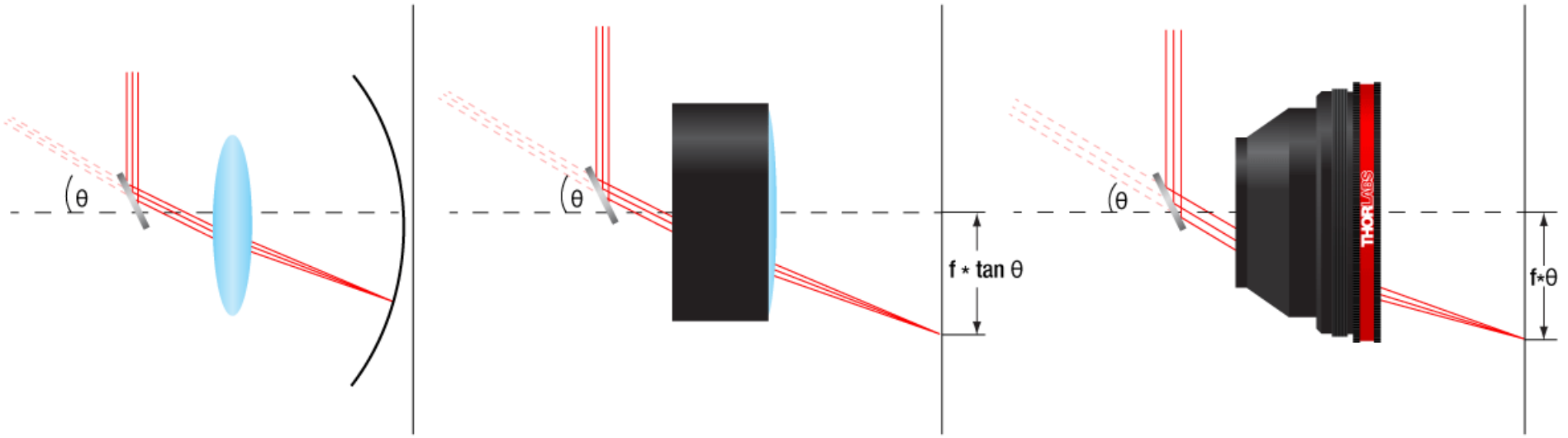


f- θ lens

linear stage movement
(up to 500 mm/sec, range: 30 cm)

galvanometer scanner
(up to 5000 mm/sec, range: 5 cm)

More to know about galvanometer scanner

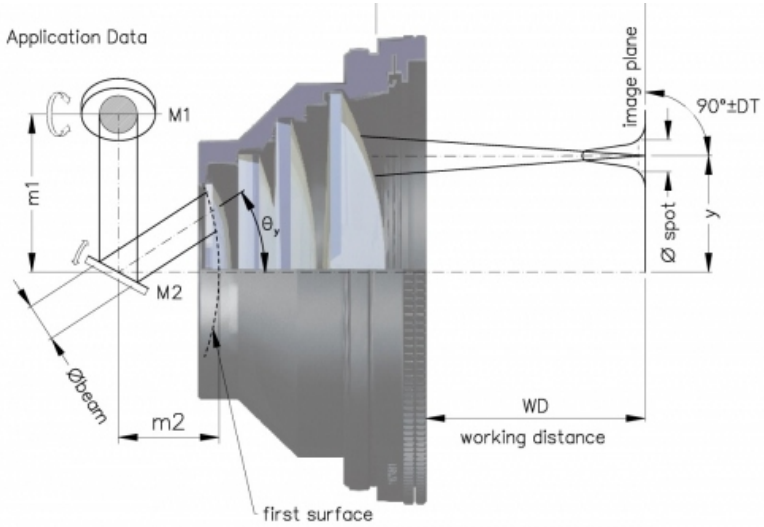


A. Spherical Lens

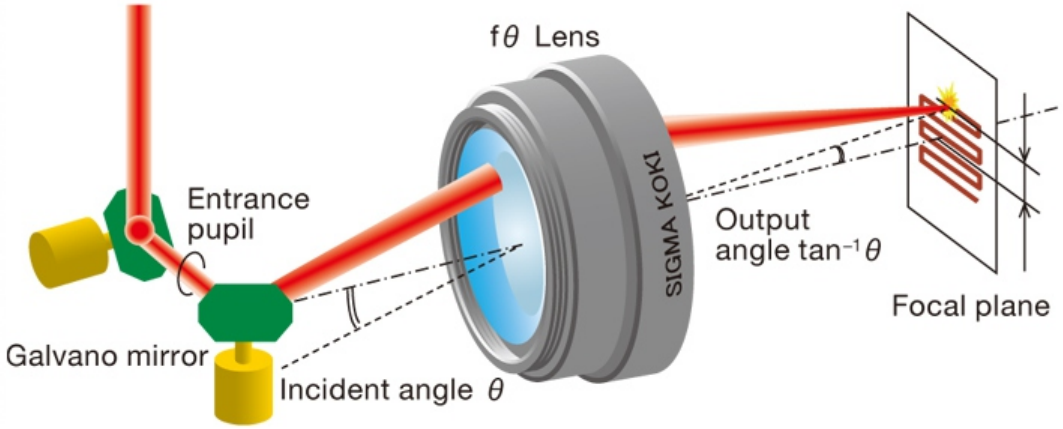
B. Flat-Field Scanning Lens

C. F-Theta Scanning Lens

http://www.thorlabs.com/

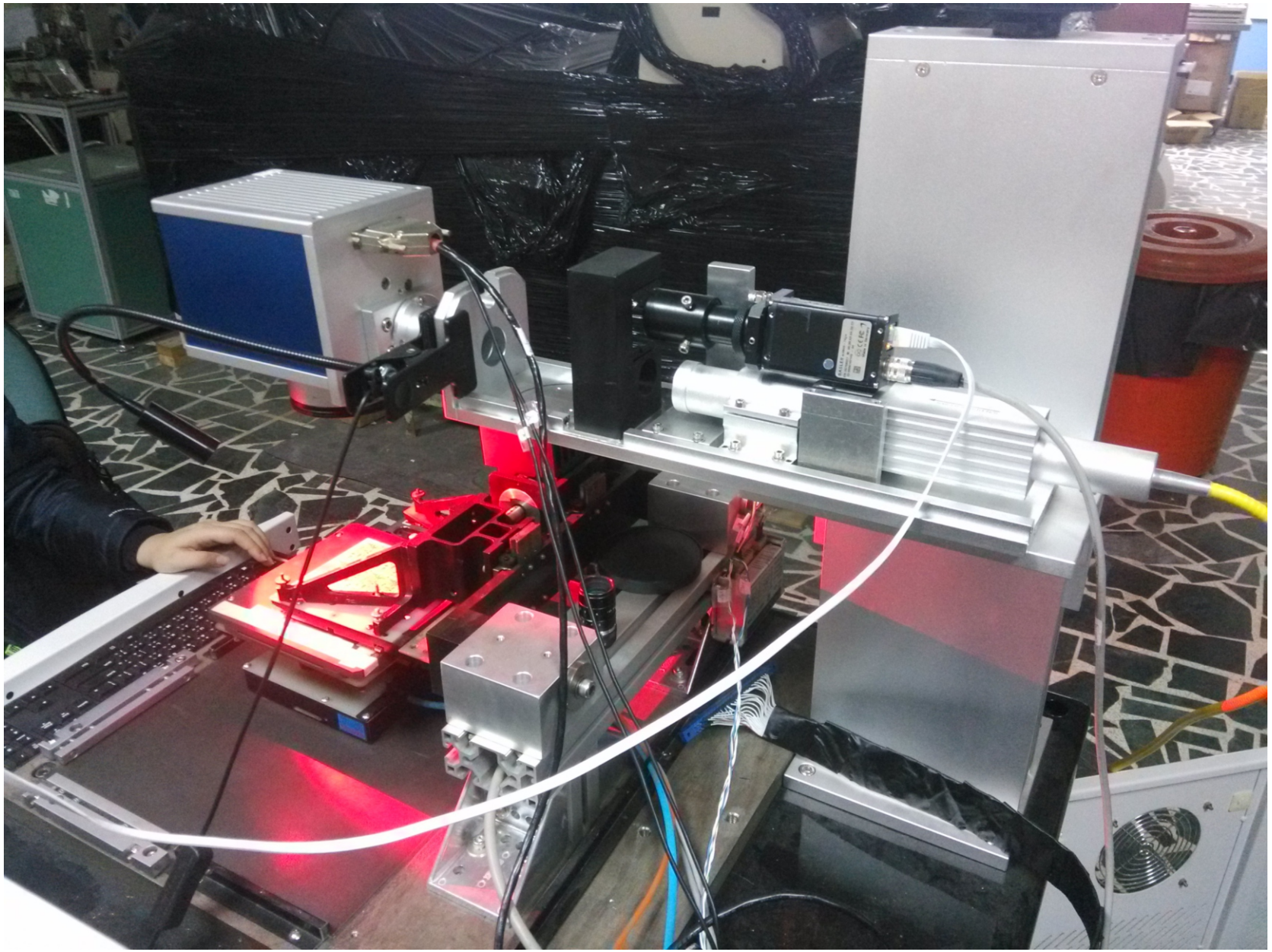


https://www.qioptiq-shop.com/



http://www.global-optosigma.com/

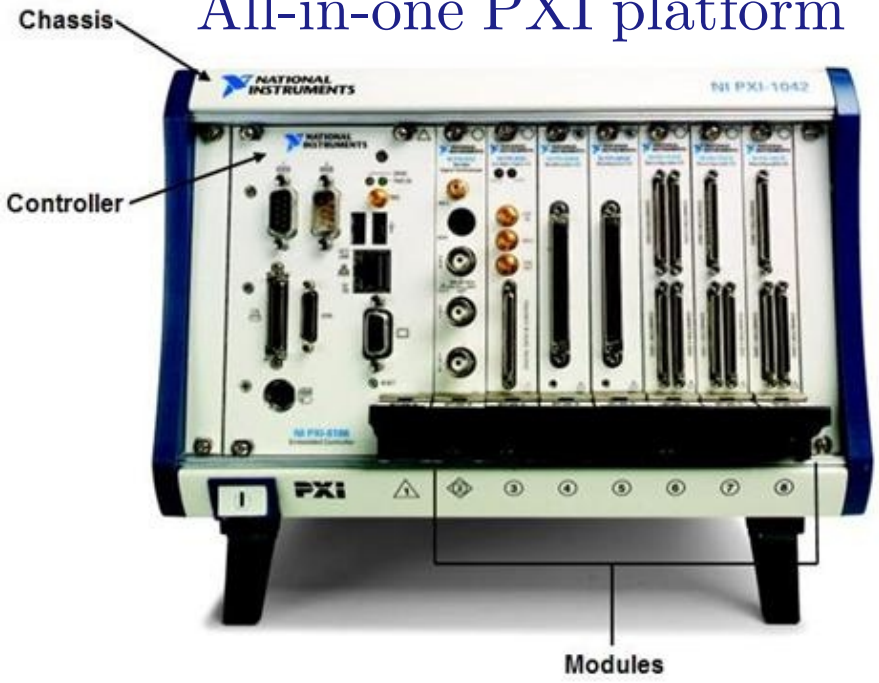
Synchronization between galvo scanner and moving stage



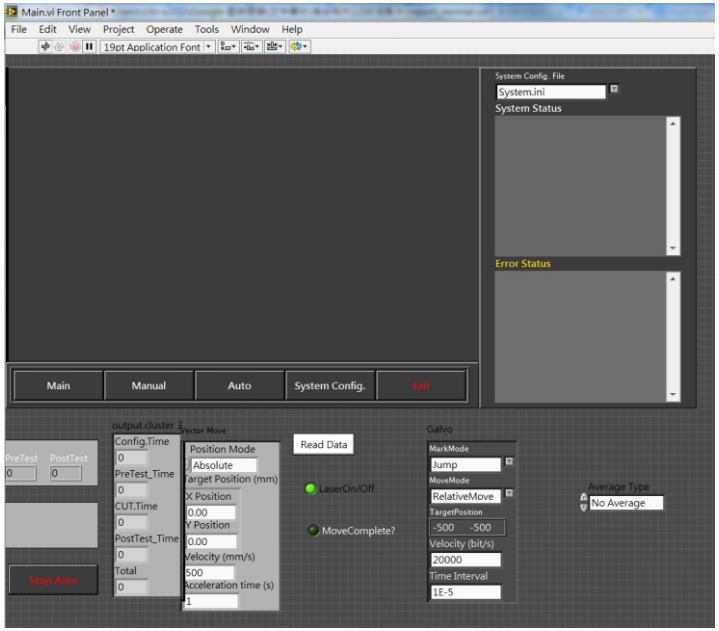
Computing platform for measurement and automation

Collaboration with National Instruments

All-in-one PXI platform



User-friendly interface design

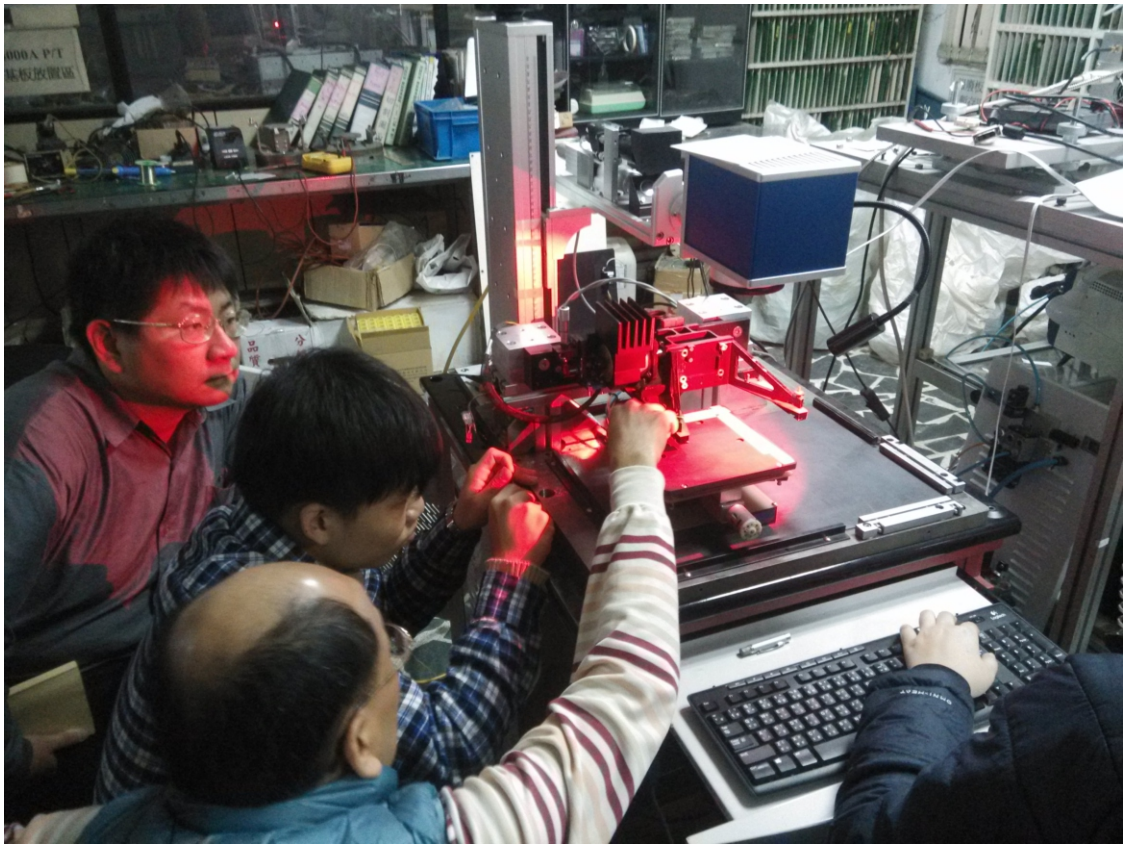


Fast sampling rate up to 750K S/s (FPGA-based) for measurements.

External GPIB equipment or DMM for ultra-high precision application.

System integration for customized Vision & Motion module.

Future works



Thank you for your attention.